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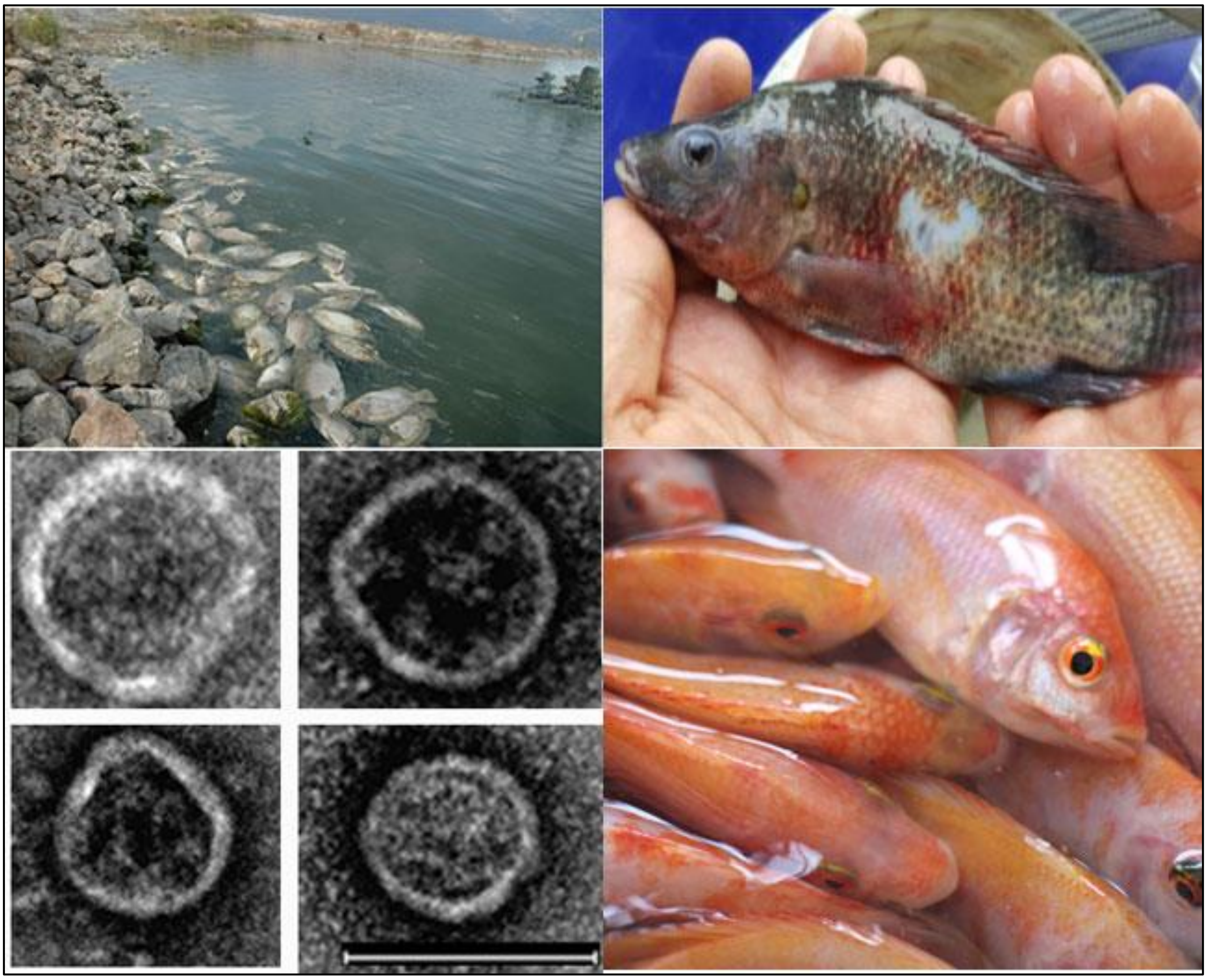
TILAPIA LAKE VIRUS DISEASE STRATEGY MANUAL



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TILAPIA LAKE VIRUS DISEASE STRATEGY MANUAL

by
Kathy F.J. Tang
Aquatic Animal Health Specialist
Tucson, United States of America

Melba G. Bondad-Reantaso
Fisheries and Aquaculture Division
Food and Agriculture Organization of the United Nations
Rome, Italy

Win Surachetpong
Associate Professor
Kasetsart University
Bangkok, Thailand

Ha Thanh Dong
Assistant Professor
Asian Institute of Technology
Bangkok, Thailand

Nihad Fejzic
Professor
University of Sarajevo
Sarajevo, Bosnia and Herzegovina

Qing Wang
Director of Laboratory of Aquatic Disease Control and Prevention
Pearl River Fisheries Research Institute
Guangzhou, People’s Republic of China

Natan Wajsbrot
Fish Health Specialist
Phibro Animal Health Corporation
Airport City, Israel

Bin Hao
Fisheries and Aquaculture Division
Food and Agriculture Organization of the United Nations
Rome, Italy

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PREPARATION OF THIS DOCUMENT

This manual provides guidance in the development of contingency plans for countries, producers, and other stakeholders in relation to outbreaks of tilapia lake virus disease (TiLVD), a viral disease affecting farmed and wild tilapia.

The document is based, in large part, on information presented and discussed by participants at the following four training courses/workshops. These are: 1) FAO/China Intensive training course on tilapia lake virus (TiLV), hosted by Sun Yat-Sen University (SYSU) in Guangzhou, China; 2) Project Inception Workshop of GCP/RAF/510/MUL: Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture, held in Nairobi, Kenya; 3) GCP/RAF/510/MUL: Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture: Intensive training course on TiLV, hosted by Kenya Marine and Fisheries Research Institute (KMFRI) in Kisumu, Kenya; 4) UTF/ZAM/077/ZAM: Technical assistance to the Zambia Aquaculture Enterprise: Training course on development of an active surveillance for epizootic ulcerative syndrome (EUS) and tilapia lake virus (TiLV) using the FAO 12-point surveillance checklist (for non-specialists) and its implementation, hosted by the University Zambia (UNZA) in Lusaka, Zambia.

The training courses and workshops were organized by the Food Safety, Nutrition and Health (NFIMF) Team of the Fisheries and Aquaculture Division of the Food and Agriculture Organization of the United Nations (FAO).

ABSTRACT

The purpose of this manual is to inform national policymakers and other stakeholders of issues related to the development of contingency plans for responding to outbreaks of tilapia lake virus disease (TiLVD), which has caused substantial mortalities, up to 90 percent, in populations of both wild and farmed tilapia in Asia, the Americas, and Africa. The causative agent for this disease is tilapia lake virus (TiLV), which infects the liver, spleen, kidney, heart, gill tissues, brain, connective tissues of muscle, and reproductive organs of tilapia. Outbreaks of TiLVD not only have devastating economic effects on producers, but also can result in a variety of socio-economic impacts on surrounding communities. It would, therefore, be prudent to implement strategies for the prevention of TiLVD and to develop contingency plans to eradicate, contain, and mitigate the impacts of the disease when outbreaks occur. This manual provides information on: 1) the nature of TiLVD; 2) diagnosis; 3) prevention and control; 4) epidemiology; 5) principles of eradication, containment and mitigation; and 6) policy development issues.

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ACKNOWLEDGEMENTS

This publication was supported by FAO projects on tilapia lake virus disease (TiLVD), namely: (i) GCP/RAF/510/MUL: Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture; and (ii) UTF/ZAM/077/ZAM: Technical assistance to the Zambia Aquaculture Enterprise. The preparation of this manual was initiated during the training courses and workshops held in China, Kenya and Zambia from 2018 to 2019, we sincerely acknowledge all the participants.

We thank the FAO regional offices in Kenya and Zambia and hosting institutions for these training courses and workshops: Sun Yat-Sen University (SYSU) of China, Kenya Marine and Fisheries Research Institute (KMFRI), and the University of Zambia.

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We thank the Norwegian Agency for Development Cooperation under the Norwegian Ministry of Foreign Affairs for support in the completion of this publication under the project GCP/GLO/979/NOR: Improving Biosecurity Governance and Legal Framework for Efficient and Sustainable Aquaculture Production and GCP/GLO/352/NOR: Responsible use of fisheries and aquaculture resources for sustainable development. The disease strategy manual is considered a ‘toolkit’ of the Progressive Management Pathway for Improving Aquaculture Biosecurity.

Special thanks are also due to Mr Sebastian Sims and Mr Paulo Padre (Food Safety, Nutrition and Health, NFIMF Team) and Ms Marianne Guyonnet (Statistics and Information, NFISI Team) for their editorial, operational and logistical support in the finalization of this document.

ABBREVIATIONS AND ACRONYMS

BPL	β-propiolactone
CA	competent authority
CD	cluster of differentiation
CPE	cytopathic effect
DNA	deoxyribonucleic acid
EKE	expert knowledge elicitation
ELISA	enzyme-linked immunosorbent assay
EUS	epizootic ulcerative syndrome
FAO	Food and Agriculture Organization of the United Nations
GCHD	grass carp haemorrhagic disease
GIFT	genetically improved farmed tilapia
GWAS	genome-wide association study
H&E	haematoxylin and eosin
HBSS	Hanks' balanced salt solution
ICTV	International Committee on Taxonomy of Viruses
iELISA	indirect enzyme-linked immunosorbent assay
IFN	interferon
IHC	immunohistochemistry
IHN	infectious haematopoietic necrosis
IL	interleukin
IP	intraperitoneal
IPNV	infectious pancreatic necrosis virus
ISAV	infectious salmon anaemia virus
ISH	*in situ* hybridization
ISKN	infectious spleen and kidney necrosis
KMFRI	Kenya Marine Fisheries Research Institute
LAMP	loop-mediated isothermal amplification
LC MS/MS	liquid chromatography-tandem mass spectrometry
MAb	monoclonal antibody
MHC	major histocompatibility complex
MMC	melanomacrophage centre
NCR	non-coding region
NFIMF	Food safety, Nutrition and Health
NFISI	Statistics and Information
nt	nucleotide
OIE	World Organisation for Animal Health
ORF	open reading frame
PB1	polymerase basic protein 1
PCR	polymerase chain reaction
PMP/AB	Progressive Management Pathway for improving Aquaculture Biosecurity
qPCR	quantitative polymerase chain reaction
QTL	quantitative trait locus
RdRp	RNA-dependent RNA polymerase
RNA	ribonucleic acid
RSIVD	red sea bream iridovirus disease
SHT	syncytial hepatitis of tilapia
SMS	summer mortality syndrome

SNP	single-nucleotide polymorphism
SPD	salmon pancreatic disease
SPR	specific-pathogen-resistant
SVCV	spring viraemia of carp virus
SYSU	Sun Yet-Sen University
$TCID_{50}$	median tissue culture infectious dose
TEM	transmission electron microscope
TiLV	tilapia lake virus
TiLVD	tilapia lake virus disease
TOMMS	tilapia one-month mortality syndrome
UNZA	University of Zambia
VHSV	viral haemorrhagic septicaemia virus
WAHIS	World Animal Health Information System

1. Introduction

Aquaculture is the most rapidly growing area of the food production sector where 52 percent of fish for human consumption are derived from (FAO, 2020a). Tilapia aquaculture production is primarily from Asia, with additional production from Africa and the Americas. In 2019, the top three producers are: China (1.6 million tonnes), Indonesia (1.3 million tonnes), and Egypt (1.1 million tonnes). Nile tilapia (*Oreochromis niloticus*) is the most predominant species (4.6 million tonnes produced in 2019) (FAO, 2021) and was first introduced to developing countries to support subsistence farming; now, large-scale commercial production has become significant and tilapia products are traded globally. Disease is one of the most serious problems for tilapia production, and these pathogens include bacteria, parasites and viruses, that are spread around the globe. These diseases have collectively cost the global tilapia aquaculture industry billions of USD. Aquaculture sectors have developed response plans to combat the threat of the diseases of aquatic animals, such as contingency plans.

Disease strategy manuals are part of technical plans, which are sets of instructions or manuals, required to support the various components of national contingency plans (Arthur *et al*., 2005). The outline of a disease strategy manual was based on the Australian Aquatic Animal Diseases Veterinary Emergency Plan (AQUAVETPLAN).

This disease strategy manual consists of the following major sections:

(1) nature of the disease (aetiology, susceptible species, global distribution);
(2) diagnosis of diseases (Levels I, II, III diagnostic methods, field diagnostic method, laboratory infection, and corroborative diagnostic criteria);
(3) prevention and treatment (fish immunity, vaccination, immunostimulant, probiotics, development of TiLV-free and TiLVD-resistant tilapia);
(4) epidemiology (geographic distribution, genotype, persistence in the environment, reservoir hosts, modes of transmission, risk factors, risk assessment, impact of TiLVD);
(5) principles of control and eradication (methods for eradication, containment and mitigation, vector control, environmental considerations, public awareness, trade and industry considerations); and
(6) policy and rationale (overall policy, problems that need to be addressed, overview of response options, improving knowledge and capability, social and economic effects, funding and compensation)

In 2009, the Israeli tilapia aquaculture industry experienced substantial production losses, and the causative agent was suspected to be a viral agent. Later, tilapia lake virus (TiLV) was isolated from affected tilapia sampled in 2011. This virus infects tilapia, including the most farmed species, Nile tilapia. This disease is important, as mortalities can reach 90 percent. Subsequent TiLVD outbreaks were reported in American, Asian and African countries. Economic losses from most cases of disease outbreaks are not available. So, the strategies to prevent substantial losses for the tilapia farming industry are important. This strategy manual provides information specific to the TiLVD, the contents consist of reviewing the existing knowledge of the disease and recommendations for its prevention and control that include measures for eradication, containment and mitigation. The manual can serve as an information base for the development of a national contingency plan for responding to TiLVD outbreaks.

2. The nature of tilapia lake virus disease (TiLVD)

TiLVD is a viral disease that has caused severe mortalities in farmed and wild populations of tilapia. Outbreaks of this disease have led to production losses to tilapia producers around the world. The disease first appeared in farmed tilapia (Figure 1) and wild *Sarotherodon galilaeus* (known as “St. Peter’s fish” by locals in Israel) in Sea of Galilee (Kinneret Lake) in Israel. Subsequent outbreaks appeared in Latin America, Africa, Asia, and most recently in North America. In Egypt, the associated mortalities occurred during the summer months, thus the disease was referred to as “summer mortality syndrome” (SMS) (Fathi *et al*., 2017). In Thailand, this disease was named “tilapia one-month mortality syndrome” (TOMMS); the mortality started 1–4 weeks after tilapia were transferred from hatcheries to grow-out facilities (Surachetpong, 2017).

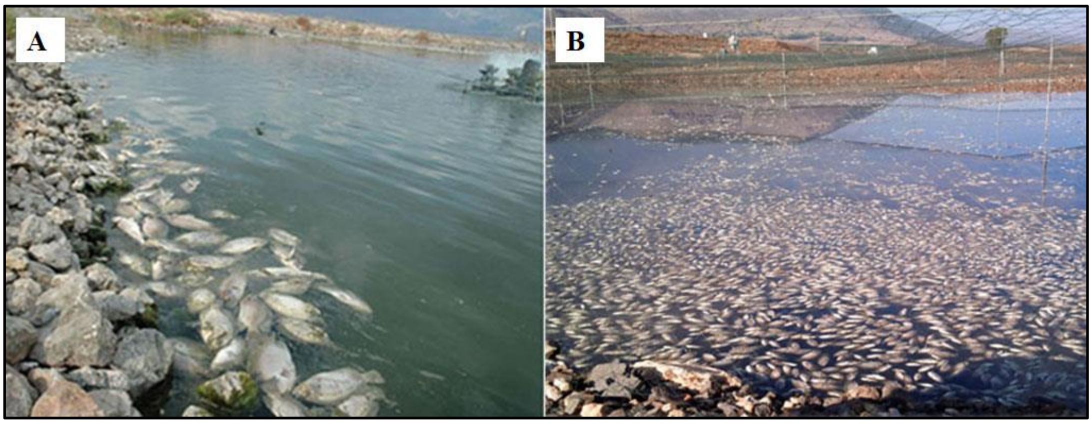

Figure 1. The occurrence of mass mortality due to TiLVD in farmed hybrid tilapia (*Oreochromis niloticus* × *O. aureus*) in Israel. Source: Eygnor *et al.*, 2014.

Through viral isolation from diseased tilapia, inoculating viral filtrate to cell-culture, and injecting viruses to naïve fish studies, the causative agent was identified as tilapia lake virus (TiLV) (Eyngor *et al*., 2014; Surachetpong *et al*., 2020). Its target tissues include liver, kidney, spleen, gills, brain, eye, and connective tissues of muscle, ovary and testis. The infected fish display clinical signs of skin redness, erosion, exophthalmos, abdominal distension, scale protrusion, and anaemia.

A syndrome known as syncytial hepatitis of tilapia (SHT) was described by Ferguson *et al* (2014). During 2011–2012, they collected, for histological evaluation, moribund tilapia from farmed populations in Ecuador that had suffered severe mortalities. These fish were characterized with tissue necrosis and syncytial cell formation in hepatocytes; the presence of virus-like particles under the transmission electron microscope (TEM) suggested its viral aetiology. Later, TiLV was isolated from these samples by Bacharach *et al.* (2016) and del-Pozo *et al.* (2017), therefore, SHT and TiLVD are the same disease.

TiLVD can cause substantial production losses in tilapia, which is a major source of low-cost, but high quality, protein for humans, and plays an important role in food security and nutrition in developing countries. The Food and Agriculture Organization of the United Nations (FAO) has disseminated TiLVD awareness information to stakeholders and issued early warning of the potential production losses from this disease (FAO, 2017a,b).

2.1 *Aetiology*

TiLV is a negative-sense, RNA virus. Under TEM, the virions are enveloped, spherical-shaped, with diameters of 55–80 nm (Figure **2**). (Eyngor *et al*., 2014; Tattiyapong, Dachavichitlead and Surachetpong, 2017). The virion has a capsid (protein shell) containing several (up to 7) electron-dense aggregates (del-Pozo *et al*., 2017). The viral morphology and size are similar to members of the family *Orthomyxoviridae* and was initially described as an orthomyxo-like virus, but its genomic sequence showed little homology to those of orthomyxoviruses.

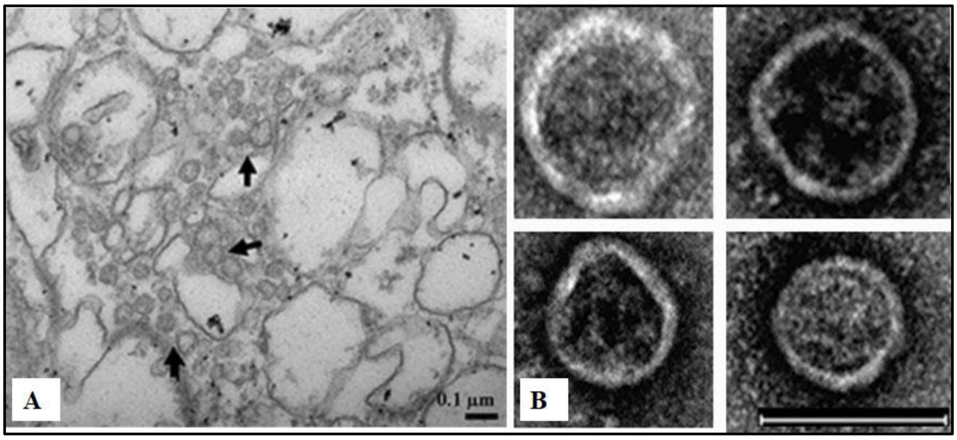

Figure 2. Transmission electron micrographs of TiLV virions. (A) Virions (arrows) are within the cytoplasmic area of a brain cell; (B) purified virions. Scale bars = 0.1 µm. Source: Tattiyapong, Dachavichitlead and Surachetpong, 2017; Eygnor *et al.*, 2014.

TiLV has recently been placed as a new species, *Tilapia tilapinevirus*, under a new family *Amnoonviridae*, within the order Articulavirales (ICTV, 2020). This virus contains ten segments of RNA, each of which consists of an open reading frame (ORF) flanked by non-coding regions (NCRs) at 5' and 3' ends. The size of each segment ranges from 456 (segment 10, smallest genomic RNA segment) to 1 641 (segment 1, largest) nucleotides (nt); the total genome size is 10 323 nt (Bacharach *et al.*, 2016, GenBank no. KU751814-KU751823). The ORF (519 amino acid) of segment 1 encodes a putative RNA-dependent RNA polymerase (RdRp) containing a conserved domain of the influenza C virus RdRp subunit, polymerase basic protein 1 (PB1). In the influenza virus, PB1 is the largest, central, subunit of the polymerase, which it binds to the conserved 5' and 3'-viral RNA termini (Fodor, Seong and Brownlee, 1993; Biswas and Nayak, 1994).

However, the amino acid sequence similarity of the putative RdRp of TiLV is only 17 percent to that of the influenza C virus (Bacharach *et al*., 2016). The ORFs from nine other segments show no similarity, thus no counterpart, to any viral proteins. All ten TiLV genomic segments have highly conserved sequences at the 3' and 5' termini, both with 13 nt. In influenza viruses, these inverted, partially, complementary termini can form a duplex structure (referred as a panhandle), which is important for the package of genomic RNAs, can also function as a promoter (a binding site for the polymerase) for viral transcription and replication (Hu *et al*., 1987).

By analysing the protein contents of the TiLV virions with liquid chromatography-tandem mass spectrometry (LC MS/MS), 9 proteins predicted from ORFs of segments 2 to 10 were identified. No protein encoded from ORF of segment 1 (putative RdRp) was found in the purified virions (Bacharach *et al*., 2016).

The TiL-4-2011 isolate, collected in 2011 in Israel by Eyngor *et al.* (2014), is the earliest characterized isolate among those studied. This isolate is used as a reference genome to assess the variations with 19 other whole genome sequenced isolates collected from Israel (2012), Ecuador (2012), Thailand (2014–2019), Bangladesh (2 isolates in 2017; 2 isolates in 2019), Peru (2018), and the United States of America (2 isolates from an outbreak in an Idaho farm in 2019) (Table 1). Despite the geographic separation, this Israeli isolate has very high nucleotide sequence identities with both Ecuadorian (98-99%) and Thai (97-99%) isolates.

Table 1. Information of whole genome sequenced TiLV isolates. Isolates were from Israel (N=2), Ecuador (N=1), Thailand (N=10), Bangladesh (N=4), Peru (N=1), United States of America (USA) (N=2).

no.	Isolate name	Country/year of collection	GenBank no.
1	TiL-4-2011	Israel/2011	KU751814-823
2	AD-2016	Israel/2012	KU552131-142*
3	EC-2012	Ecuador/2012	MK392372-381
4	TH-2014	Thailand/2014	MN687695-704
5	TH-2015	Thailand/2015	MN687705-714
6	TV1	Thailand/2016	KX631921–930
7	TH-2016-CU	Thailand/2016	MN687715-724
8	TH-2016-CN	Thailand/2016	MN687725-734
9	BD-2017	Bangladesh/2017	MN939372-381
10	BD-2017-181	Bangladesh/2017	MT466437-446
11	TH-2017	Thailand/2017	MN687735-744
12	F3-4	Peru/2018	MK425010-019
13	WVL18053-01A	Thailand/2018	MH319378-387
14	TH-2018-N	Thailand/2018	MN687745-754
15	TH-2018-K	Thailand/2018	MN687755-764
16	WVL19031-01A	USA/2019	MN193513-522
17	WVL19054	USA/2019	MN193523- 532
18	TH-2019	Thailand/2019	MN687765-774
19	BD-2019E1	Bangladesh/2019	MT466447-456
20	BD-2019-E3	Bangladesh/2019	MT466457-466

*: AD-2016 consists of 12 instead of 10 segment sequences; both KU552135 and KU552139 are parts of viral segment-3; KU552136 and KU552141 are parts of viral segment-5.

Thawornwattana *et al*. (2020) performed Bayesian phylogenetic analyses of the whole genome sequences of 17 isolates from 6 countries and estimated TiLV evolution rates of 1.81–3.47 x 10^{-3} substitutions per site per year. They estimated the virus to have originated between 2003 and 2009, in agreement with that suspected TiLVD-associated mortality was first noticed in 2009 (Eyngor *et al*., 2014). TiLV started to spread and reached its peak in 2014–2016. From 2016 onward, global TiLV populations declined steadily. This could be a result of the use of TiLVD-resistant tilapia and/or a better and more cautious protocol of tilapia importation.

The whole genome phylogenetic analyses also revealed reassortment of TiLV in 3 isolates: Israel isolate Til-4-2011 contains EC-2012's segments 5 and 6; TH-2018-K consists of BD-2017's segment 5; TH-2016-CN contains the segments 1–4 from an unknown TiLV isolate (Chaput *et al*., 2020; Thawornwattana *et al*., 2020). The reassortment is from co-infection of a host cell by two (or more) strains and results in shuffling viral segments in the progeny viruses; this event may increase TiLV genetic diversity.

2.2 ***Susceptible species***

TiLV is known to infect tilapia, which is the common name applied to three genera of fish in the family Cichlidae: *Oreochromis*, *Sarotherodon* and *Tilapia*. The susceptible farmed tilapia include Nile tilapia, gray tilapia, red tilapia, red hybrid tilapia and Mossambique tilapia (Table 2). TiLV has also been detected in wild *Sarotherodon galilaeus* (St. Peter's fish) in the Sea of Galilee (Kinneret Lake) (Eyngor *et al*., 2014). Although TiLV can cause disease in all life stage of susceptible fish, the small fish are more susceptible to TiLV infection than large fish (Roy *et al*., 2021).

TiLV has a narrow host range, and mainly infects tilapia fishes in the farm and natural environments. However, Jaemwimol *et al*. (2018) showed that giant gourami (*Osphronemus goramy*) can be infected by TiLV through laboratory infection studies. The infected fish developed clinical signs and showed cumulative mortality of 60–100 percent from the challenge bioassays (Table 2). TiLV-infected giant gourami were also detected in the farm environment; these fish were cohabitating with TiLV-infected Nile tilapia in the same water body but in separate nets (Chiamkunakorn *et al*., 2019). Laboratory challenge experiments also revealed that zebrafish (*Danio rerio*) are susceptible to TiLV (Rakus *et al*., 2020; Widziolek *et al*., 2021). Ornamental African cichlid *Aulonocara* spp. were shown to be susceptible to TiLV in experimental challenge studies, injected fish had a cumulative mortality of 56.25% (Yamkasem *et al*., 2021a).

Table 2. Species susceptible to tilapia lake virus infection.

Common (scientific) name	**Infection type**	**Mortality**	**Reference**
Nile tilapia (*O. niloticus*)	Natural and experimental	Yes	Eyngor *et al*., 2014; Ferguson *et al*., 2014; Del-Pozo et al., 2016; Dong *et al*. 2017a; Fathi *et al*., 2017; Surachetpong *et al*., 2017
Gray tilapia (*O. niloticus* × *O. aureus*)	Natural and experimental	Yes	Eyngor *et al*., 2014; Tsofack et al., 2017**;** Mugimba, et al., 2019
Red tilapia (*Oreochromis* spp.)	Natural and experimental	Yes	Dong *et al*., 2017a; Surachetpong *et al*., 2017; Mugimba *et al*., 2019; Nicholson *et al*., 2020
Red hybrid tilapia (*O. niloticus* x *O. mossambicus*)	Natural	Yes	Amal *et al*., 2018
Mozambique tilapia (*O. mossambicus*)	Experimental	Yes	Waiyamitra et al., 2021
Mango tilapia (*Sarotherodon galilaeus*)	Natural	Yes	Eyngor *et al*., 2014
Giant gourami (*Osphronemus goramy*)	Experimental and natural	Yes	Yamkasem et al., 2018; Chiamkunakorn *et al*., 2019
Zebra fish (*Danio rerio*)	Experimental	Yes	Rakus *et al*., 2020; Widziolek et al., 2021
Malawi cichlid (*Aulonocara* spp.)	Experimental	Yes	Yamkasem et al., 2021

Although TiLV can cause disease in all life stage of susceptible tilapia, the small fish are more susceptible to TiLV infection than large fish (Roy *et al*., 2021).

2.3 *Global distribution*

TiLV was first recognized in Israel in 2011 (Eyngor *et al*., 2014; Bacharach *et al.*, 2016). Subsequently, the disease was reported in several other countries: Ecuador (year not known, first viral isolate was collected in 2012) (Ferguson *et al*., 2014; del-Pozo *et al*., 2017; Tsofack *et al*., 2017); Colombia (year not known; Tsofack *et al*., 2017; Contreras *et al*., 2021); Thailand (2015) (Dong *et al*., 2017a,b; OIE, 2017a; Surachetpong *et al*., 2017); Egypt (2015) (Fathi *et al*., 2017; Nicholson *et al*., 2017); United Republic of Tanzania (2015); Uganda (2016) (Mugimba *et al*., 2018); India (2016) (Behera *et al*., 2018); Indonesia (2016) (Koesharyani *et al*., 2018); Taiwan Province of China (2017) (OIE, 2017b); the Philippines (2017) (OIE, 2017c); Malaysia (2017) (OIE, 2017d; Amal *et al*., 2018); Bangladesh (2017) (Chaput *et al*., 2020; Debnath *et al*., 2020); Peru (2017) (OIE, 2018a; Pulido *et al*., 2019); Mexico (2018) (OIE, 2018b); and most recently in the USA (2019) (Ahasan *et al*. 2020) (Figure 3).

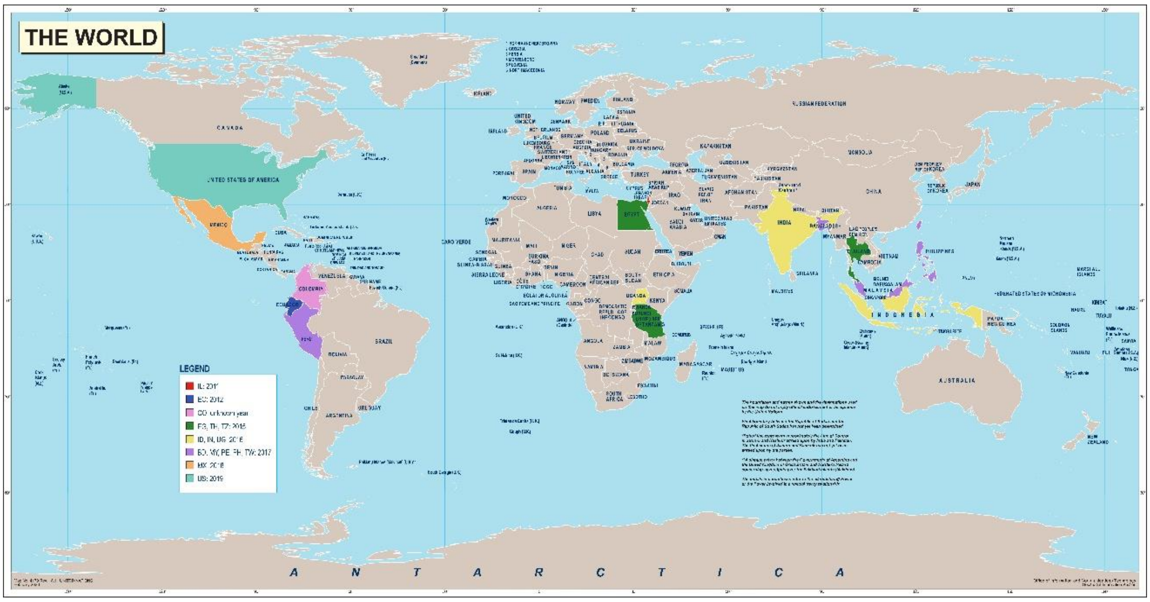

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Figure 3. The detection of TiLV in 16 countries/province and the year of first report. These are Israel (IL, 2011), Ecuador (EC, 2012), Colombia (CO, unknown year), Egypt (EG, 2015), Thailand (TH, 2015), United Republic of Tanzania(TZ, 2015), Indonesia (ID, 2016), India (IN, 2016), Uganda (UG, 2016), Bangladesh (BD, 2017), Malaysia (MY, 2017), Peru (PE, 2017), the Philippines (PH, 2017), Taiwan Province of China (TW, 2017), Mexico (MX, 2018), USA (US, 2019). Map No. 4170 Rev. 19, United Nations, October 2020.

Dong *et al*. (2017b) used TiLV RT-PCR to screen tilapia samples, archived and fresh, from four hatcheries in Thailand with records of international trade of live fish. Most of these samples, which were collected during the period of 2012 to 2017, tested positive. Since many countries import tilapia fry and fingerlings from these hatcheries, it is possible, although no specific instances have been documented, that TiLV could have been spread to other countries through exports of fry or fingerlings. A lack of comprehensive investigation of mortality-associated incidents in tilapia could mean that the geographic distribution of TiLV may be wider than currently known. For example, a report of substantial mortality in tilapia in Zambia in 2016 was not well investigated to determine if TiLV was the causative agent.

3. Diagnosis of disease

Infected fish had a high mortality, up to 90 percent, and exhibited gross signs such as skin erosion, haemorrhage, and abdominal distension (see below). Examination of histological sections of liver, brain and other internal organs, stained with haematoxylin and eosin (H&E) can provide a presumptive diagnosis of TiLV infection. However, for definitive diagnosis, and to determine the infection status in sub-clinically infected fish, virus isolation/cell-culture and RT-PCR (or quantitative reverse transcription polymerase chain reaction; these tests have a fast turnaround time) testing are recommended in the OIE disease card (OIE, 2020). In addition, the laboratory virus infection to naïve fish can be used for the confirmation of TiLV status. Therefore, the diagnosis can depend on the gross signs, histopathology, virus isolation/cell-culture, RT-PCR-based methods and laboratory infection study. A flow chart of the diagnosis of TiLVD is given in Figure 4.

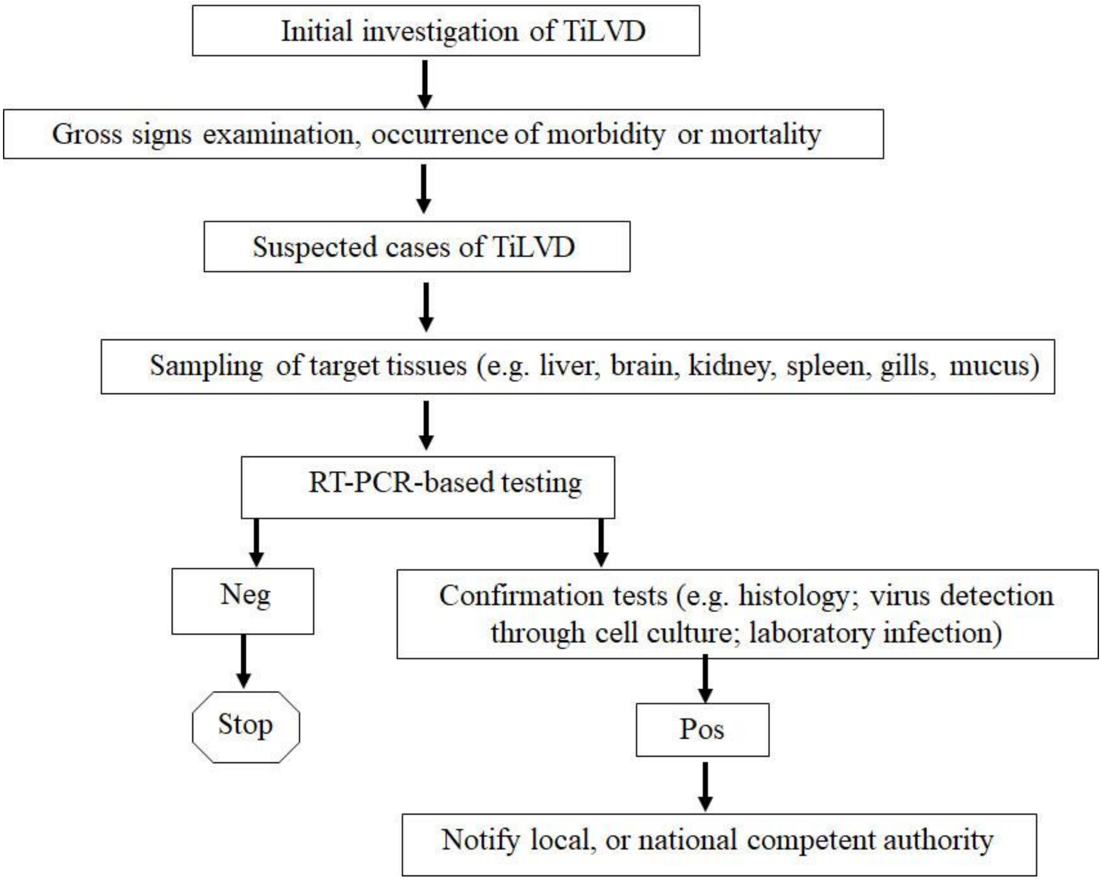

Figure 4. TiLVD diagnostic flowchart. Pos: positive, Neg: negative.

The following sections describe these means of diagnosis: observation of clinical signs, histopathology, RT-PCR-based and RT-loop-mediated isothermal amplification (LAMP) methods, virus isolation and cell-culture systems, laboratory infection study to verify virus isolates' pathogenicity. The diagnostic methods of aquatic animals can be categorized into three levels as described by Bondad-Reantaso *et al.* (2001):

Level I: examination of gross signs and observations of animals' behaviours

Level II: isolation and examination of pathogens in parasitology, bacteriology and mycology and histopathological evaluations of infected hosts;

Level III: virus isolation, cultivation in the cell culture system, TEM examination, and molecular techniques (PCR-based, antibody-based assays).

3.1 *Gross signs (Level I)*

The onset of gross signs and mortality usually starts within one month after stocking the fingerlings into the grow-out facilities. The gross signs may include haemorrhage at the base of fins and opercula, skin erosion, scale protrusion and loss (Figure 5A and 5B), abdominal swelling due to the presence of ascites (Figure 5B), skin darkening, gill pallor, and ocular alterations such as exophthalmia ("pop-eye"), shrinkage of the eyeball (phthisis bulbi) in severe infection (Figure 5C) and opacity of the lens. Fish also exhibit abnormal behaviour like lethargy, loss of appetite, swimming at the surface, and loss of balance (Eyngor *et al*., 2014; Ferguson *et al*., 2014; Dong *et al*., 2017a; Surachetpong *et al*., 2017; Tattiyapong**,** Sirikanchana and Surachetpong, 2018). TiLVD associated mortalities may range from 10 to 90 percent.

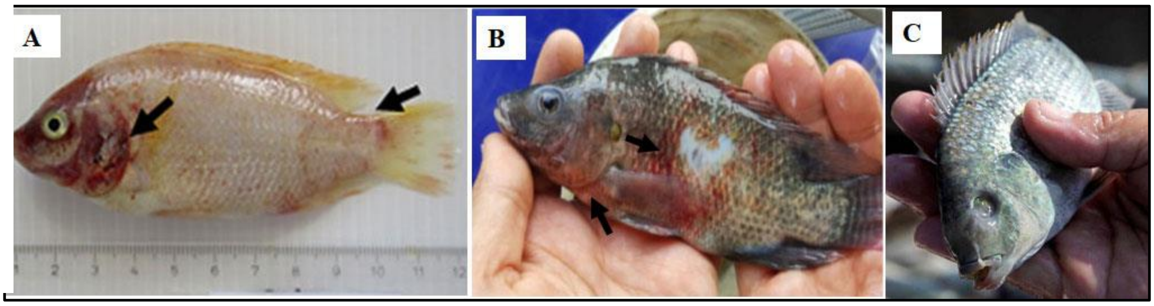

Figure 5. External gross signs of TiLVD in clinical samples. (A) diseased red tilapia showed haemorrhage (black arrows); (B) diseased Nile tilapia showed skin erosion, haemorrhage on various parts of body, loss of scales, abdominal swelling, and protruding eyeball (exophthalmos); (C) diseased wild tilapia (*Sarotherodon galilaeus*) showed shrinkage of the eye and loss of ocular functioning (phthisis bulbi). Source: Tattiyapong, Dachavichitlead and Surachetpong, 2017; Eygnor *et al.*, 2014.

Internally, fish infected with TiLV showed some abnormalities such as pale, necrotic and watery liver. In some cases, the liver tissue turned green or dark color (Figure 6). Infected fish also showed enlarged spleen and gall bladder, accumulation of fluid in the intestine and abdomen cavity.

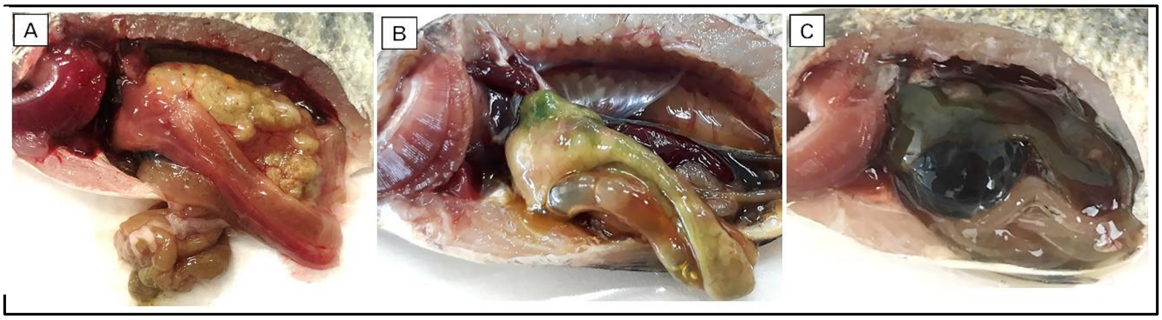

Figure 6. Internal organs: normal and TiLV-infected tilapia. (A) normal fish, (B) fish from TiLVD outbreak, (C) experimentally infected fish.

3.2 *Field diagnostic methods*

Field diagnostic systems should be rapid, inexpensive, easy to maintain and operate for non-specialists. In addition, the reagents should be provided in a format that allows easy shipping and storage. There is a commercial kit currently available for field detection of TiLV. The analyser is called POCKIT™ (manufactured by GeneReach Biotechnology Corp.) and has been certified by OIE. The assay can be completed within 45 minutes. A hand-held device (POCKIT™ Micro) is also available for pond-site detection of TiLV.

3.3 *Laboratory methods*

3.3.1 Sample submission

At the first sign of the disease, clinical samples (including liver, spleen, kidney, heart, brain, gills, whole fry; see 3) should be sent to local (regional or state) aquatic animal diagnostic laboratories. If TiLVD is diagnosed to be positive, the Competent Authority (CA) should be notified, and the diagnosis confirmed through testing with other methods and/or testing additional samples from the suspected TiLV-infected populations. The CA should be contacted directly to obtain information on what additional clinical specimens may be required, and how they should be collected, stored, and transported to satisfy requirements for confirming the original diagnosis of TiLVD.

Table 3. Fish tissues to be used for TiLV testing.

Fish stage	Tissues to be sampled
Yolk sac fry	remove the yolk sac, sample the whole fry
Fish 4–6 cm	entire viscera including the kidney
Fish >6 cm to adult	liver, kidney, spleen, heart, gills, brain, muscle, blood, mucus from the body surface

The number of individual samples required when screening for TiLV will depend on the sensitivity and specificity of the diagnostic protocol, the population size, the disease prevalence, and the level of confidence desired. The laboratory procedures can follow the recommendations described in the OIE disease card (OIE, 2020).

Based on a diagnostic test with a 100 percent in both sensitivity and specificity, the sample size needed to demonstrate TiLV-free status would be determined by assuming a 2 percent prevalence, the number of individual fish that should be collected for diagnosis is 149 (OIE Aquatic code, OIE 2019). If gross signs and/or mortalities are observed, sampling 5 fish should be sufficient for each epidemiological unit from a diseased population. For detecting sub-clinically infected fish or for targeted surveillance where many samples are required, tissues at similar weights or volumes from each fish (e.g. 5 fish) can be pooled into one sample (Yamkasem *et al*., 2021b).

It is best to split a tissue sample into 3 parts (histology, RT-PCR and viral isolation). One part is fixed in 10 percent buffered (such as phosphate buffered saline [PBS]) formalin for histological evaluation, another is frozen (at -20 ^{o}C or -80 ^{o}C) and the third is preserved in 95 percent ethanol (or RNA*later*, or equivalent reagents). Frozen or ethanol-preserved tissues can be used for RT-PCR (or RT-qPCR) analyses. Rawiwan *et al*., (2021) showed that different preservation methods and storage conditions could affect the quality and amount of TiLV genomic RNA. Indeed, RNAlater® and deep-freezing at -20^{o}C offered the best storage conditions to maintain TiLV genomic RNA. The tissues for viral isolation should be aseptically removed and immersed in Hanks' balanced salt solution (HBSS) (see section 3.3.6), or immediately frozen at -80 ^{o}C, or stored on dry ice. The quality of the specimens is important and must be properly stored and transported to avoid RNA degradation and/or virus inactivation. For longer periods of storage, the fish samples should be stored at -80 °C.

3.3.2 **Non-lethal sampling**

Body mucus from infected fish can be used as a source for a non-invasive sampling protocol (Liamnimitr *et al*., 2018). In addition to being analysed by RT-PCR (or RT-qPCR), the viruses in mucus are viable and can result in a cytopathic effect (CPE) in cell culture systems (see below). The procedure involves collecting mucus (approximately 200 µL) from the fish body using a thick cover glass (or microscope slide) and scraping the skin along the lateral line from the anterior to posterior direction. This procedure can also be used for pond-site diagnosis and combined with a portable diagnostic device, after which the sampled fish can be returned to the ponds (or tanks) in the farms.

Blood can be used for detecting TiLV in subclinically infected fish (Chiamkunakorn *et al*., 2019). Before collecting blood, fish should be anaesthetized using tricaine methanesulfonate (MS-222, 50–100 ppm; or a similar anaesthetic agent). Anaesthetized fish should not react to stimuli or handling. A 1 mL syringe with a 22G needle can be inserted into the caudal vein to withdraw ~100 µL of blood, which can be transferred into a microcentrifuge tube, and add 1 mL of RNA extraction reagent, such as TRIzol. The TRIzol-added blood sample can then proceed to RNA extraction or stored at -20 °C until use. After blood sampling, fish can be placed into clean water with proper aeration to recover.

3.3.3 **Histopathology (Level II)**

For histopathology, moribund fish can be euthanised by MS-222 or clove oil (eugenol solution, 20–100 ppm) The target tissues are dissected, fixed in 10 percent buffered formalin for 24–48 hours (depending on the fish size), and transferred to 70 percent ethanol for storage. Then, fixed tissues can be processed into paraffin blocks, sectioned, and stained with H&E using a standard protocol. Stained sections are examined under a microscope. Histological examinations usually reveal lesions in the liver, spleen, kidney, and brain. The normal tilapia histology has been published by Morrison, Fitzsimmons and Wright Jr (2006), which can be used for comparison to identify histological features associated with TiLVD.

In the **liver**, there are syncytial cells (cells contains multiple nuclei, ranging from 3–4 nuclei up to 15–20 nuclei per cell) formation (Figure 7A), which is the reason that TiLVD was also named as syncytial hepatitis of tilapia (SHT) by Ferguson *et al*. (2014) and by del-Pozo *et al*. (2017). However, the formation of syncytial cells can be found in herpesviruses infected turbot (*Scophthalmus maximus*) and rainbow trout (*Oncorhynchus mykiss*) (Wolf, 1988), and reo-like virus infected halibut (*Hippoglossus hippoglossus*) (Cusack *et al*., 2001; Ferguson, Millar and Kibenge, 2003). In the TiLV-infected tilapia, massive cellular necrosis with pyknotic and karyolytic nuclei are also found in the hepatocytes, along with eosinophilic cytoplasmic inclusion bodies (Figure 7B). Severe infiltrating lymphocytes, some of them are near veins (perivenular lymphocytes) are noted (Senapin *et al*., 2018).

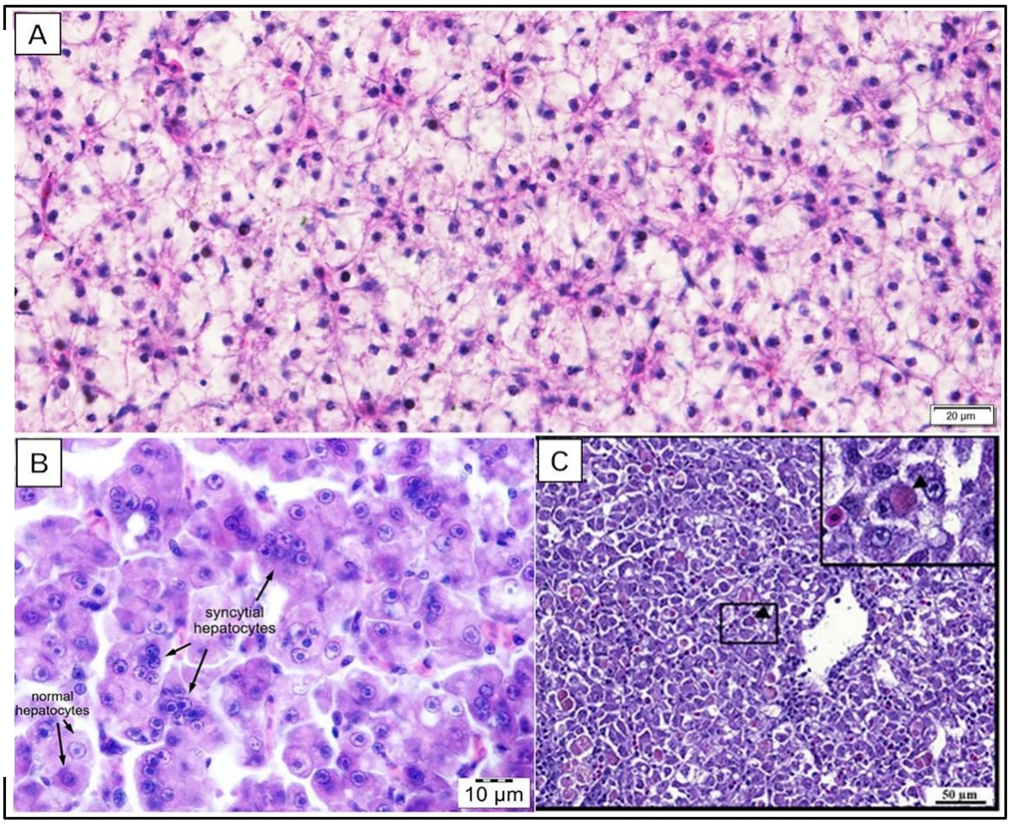

Figure 7. Histology of normal and histopathology of TILV-infected liver in tilapia. (A) normal liver; (B) typical lesion of syncytial hepatitis; (c) extensive hepatocellular necrosis with eosinophilic intracytoplasmic inclusion body in liver cells (insert: higher magnification).

In the **spleen**, the size and number of melanomacrophage centres (MMC, known as macrophage aggregates) are increased, and eosinophilic cytoplasmic inclusion bodies are present (Figure 8A).

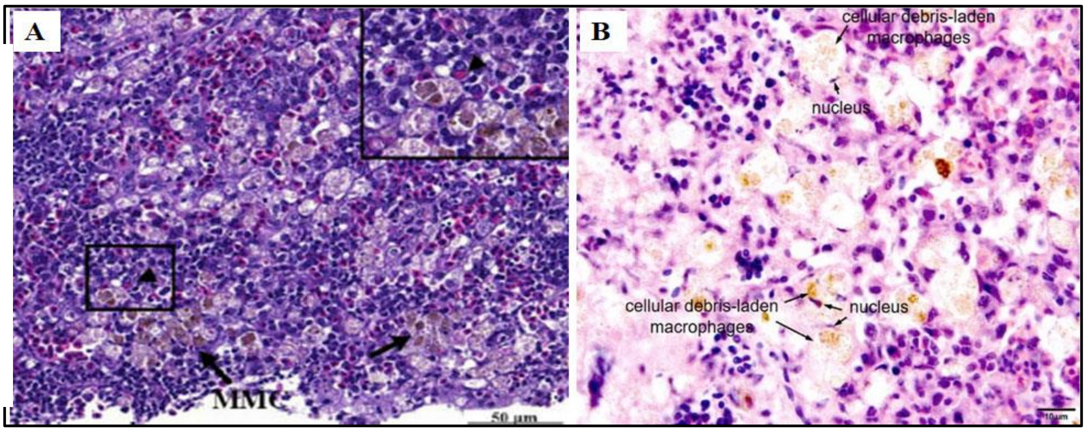

Figure 8. Histopathology of spleen in TiLV-infected tilapia. (A) Increased melanomacrophage centre (MMC) (arrows) with eosinophilic intracytoplasmic inclusion body (arrowhead in the boxed area), bar = 50 µm; (B) cellular debris-laden macrophages, bar = 10 µm.

MMCs are normally located in the stroma of the haematopoietic tissue of the spleen and the kidney, its presence is associated with late stages of chronic infection as a response to pathogens' infections (especially viruses) or poor environmental conditions. Debris-laden macrophages were present in the spleen; these macrophages were gathering at sites of infection and cleaning up debris from viral infection (Figure 8B). In the **kidney**, multiple necrotic foci were observed (Figure 9).

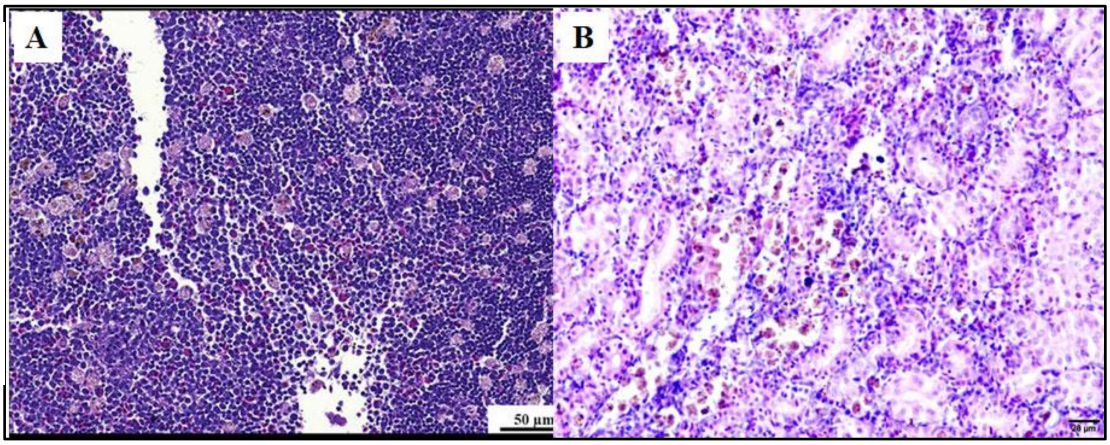

Figure 9. Histopathology of kidney in TiLV-infected tilapia. (A) Accumulation of eosinophilic proteinaceous contents, bar = 50 µm; (B) multifocal area of necrosis and inflammatory cells infiltration, bar = 20 µm.

The lesions of the **brain** include oedema, focal haemorrhages in the leptomeninges, congestion of blood vessels (Figure 10A), and congestion in both the white and grey matter; perivascular cuffs of lymphocytes are detected (Figure 10B). Some neurons within the telencephalon, particularly in the optic lobes, display various levels of degeneration (Eyngor *et al*. 2014; Tattiyapong, Dachavichitlead and Surachetpong, 2017).

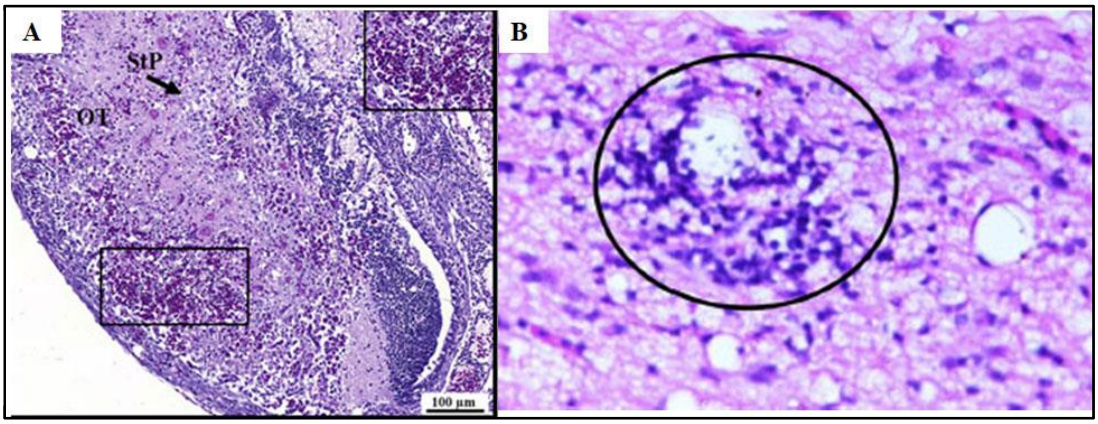

Figure 10. Histopathology of brain in TiLV-infected tilapia. (A) Multifocal haemorrhages with severe blood congestion in the brain, bar = 100 µm; (B) perivascular cuffs of lymphocytes (encircled). Source: Eygnor *et al*., 2014.

Although histopathology can provide a presumptive diagnosis of TiLVD, confirmatory tests should be performed. These include viral isolation through cell culture systems, nucleic acid amplification-based methods (such as RT-PCR, RT-LAMP), *in situ* hybridization (ISH), immunodetection, transmission electron microscopy (TEM) examination, and laboratory infection to naïve fish.

3.3.4 **Transmission electron microscopy (TEM) (Level III)**

Diagnosis of TiLV can be confirmed by TEM, which reveals the presence of spherical virus particles in the cytoplasmic areas (see Figure **2**). Target tissues (e.g. liver, spleen) should be divided into two parts. One part will be processed for histological examination, the other part will be fixed in 2 percent glutaraldehyde and stored at 4 °C. If the histological evaluation reveals lesions and/or the presence of inclusion bodies, then further investigation by TEM is warranted. The glutaraldehyde-fixed sample can be contrasted with 1 percent osmium tetroxide, resin embedded, ultrathin sectioned and examined under TEM.

3.3.5 **Molecular techniques (Level III)**

(a) RT-PCR-based method

Several RT-PCR and RT-qPCR protocols have been described for the specific and sensitive detection of TiLV in samples (Eyngor *et al*., 2014; Dong *et al*., 2017a; Tsofack *et al*., 2017; Mugimba *et al*., 2018; Nicholson, Rawiwan and Surachetpong, 2018; Tattiyapong, Sirikanchana and Surachetpong, 2018; Waiyamitra *et al*., 2018; Taengphu *et al*., 2020). Some of these methods are more sensitive and can be used to detect asymptomatic infection in apparently healthy tilapia (Dong *et al*., 2017a; Mugimba *et al*., 2018; Senapin *et al*., 2018; Taengphu *et al*., 2020). All of the current PCR protocols show the capability to detect TiLV in fish tissues with different sensitivity and specificity (Yamkasem, Tattiyapong and Surachetpong, 2020). Details on how to perform these tests can be found in the original publications and in the OIE Technical Disease Card for TiLV (OIE. 2020). Commercial RT-PCR and RT-qPCR kits for detection of TiLV are also available, for example, IQ REAL TiLV Quantitative System and the IQ Plus TiLV kit for pond-site diagnosis (GeneReach Biotechnology Corp.).

(b) Reverse transcription loop-mediated isothermal amplification (RT-LAMP)

LAMP-based amplification has been shown to be a specific and sensitive assay for detecting various pathogens (Notomi *et al*. 2000). The target fragments are amplified using 4–6 specific primers and Bst DNA (deoxyribonucleic acid) polymerase at an isothermal condition. A few TiLV RT-LAMP assays have been developed; the amplified products were visualized by the appearance of ladder-like bands after gel electrophoresis, by the observation of colour changes with naked eye, or by the emission of fluorescence under UV light (Phusantisampan *et al*., 2019; Yin *et al*., 2019; Phusantisampan *et al*., 2020). This system allows detecting TiLV by the pond-site with the use of low-cost equipment (e.g. water bath) and simple detection methods.

(c) *In situ* hybridization (ISH)

In situ hybridization using TiLV-specific probe indicates the presence of virions in the infected tissues. Positive reactions were observed in liver, brain, kidney, gills, spleen, and connective tissue in the muscle of infected fish and reproductive organs of broodstock and fertilized eggs (Bacharach *et al*., 2016; Dong *et al*., 2017a; Yamkasem *et al*., 2019; Dong *et al*., 2020). Recently, Dinh-Hung *et al.* (2021) reveled the in-depth localization of the virus in the central nervous system with the strongest positive signals in the forebrain (responsible for learning, appetitive behaviour and attention) and the hindbrain (involved in controlling locomotion and basal physiology). The result also suggests that the virus may productively enter into the brain through the circulatory system and widen broad regions through the cerebrospinal fluid along the ventricles, and subsequently induce the brain dysfunction.

(d) Immunodetection

An indirect enzyme-linked immunosorbent assay (iELISA) has been developed for detecting TiLV-elicited IgM in infected tilapia (Hu *et al*., 2020). This assay, after optimization, was demonstrated to have diagnostic sensitivity and specificity of 80.8 percent and 95.6 percent respectively, when compared to the results obtained using semi-nested RT-PCR developed by Dong *et al*. (2017a). iELISA will be useful for immunological surveillance and determining the effectiveness of vaccination. Recently, an immunohistochemistry (IHC) using rabbit derived-TiLV IgG antibody revealed spatial cellular distribution of TiLV antigens in infected tissues including spleen, liver, kidney, gills, intestines, brain, as well as endothelial cells and circulating leukocytes, suggesting the endotheliotropism and lymphotropism of this virus (Piewbang *et al*., 2021).

A rabbit polyclonal antibody against TiLV was generated and used to detect TiLV through an immunohistochemistry (IHC) procedure (Piewbang *et al*., 2021). The location and intensity of the IHC signals corresponded with the TiLV viral loads. This assay was also used to detect TiLV infection in the experimentally challenged ornamental cichlids, Mozambique tilapia, giant gourami, and naturally infected Nile tilapia.

3.3.6 **Cell-culture system (Level III)**

The presence of TiLV can be determined by isolating the viruses from the suspect fish with a standard procedure (Eyngor *et al*., 2014; Liamnimitr *et al*., 2018). The fresh or frozen tissues (blood, mucus, liver, or other target tissues) should be aseptically sampled and transport them to laboratory by placing them in 10 percent HBSS at a ratio of one volume of tissues in at least five volumes of HBSS with added antibiotics to suppress the growth of bacterial contaminants. In the laboratory, the tissues will be minced, homogenized (in 10% HBSS) and then centrifuged at 3 000 ×g for 10 min. The supernatant should be filtered through a 0.22 µm membrane, the filtered homogenates should be stored at -80 ^{o}C.

The filtered virion-containing homogenate will be inoculated in continuous cell lines such as E-11 (derived from snakehead fish *Channa striata*), CFF (from Malayan leaffish *Pristolepis fasciatus*), OmB and TmB (from *Oreochromis mossambicus*), *Onl*B (from Nile tilapia brain), *Onl*L (from Nile tilapia liver), or CAMB (from hybrid snakehead brain), then incubated at 25–27 ^{o}C allowing for viral replication. After inoculation, the cultured cells are observed daily for the appearance of CPE. Infected cells display shrinkage, aggregation, rounding up and detach from the culture surface (Figure 1). Usually, TiLV-positive samples will induce CPE in the cell lines after 4–12 days (Eyngor *et al*., 2014; Tsofack *et al*., 2017; Behera *et al*., 2018; Thangaraj *et al*., 2018; Wang *et al*., 2018, 2020).

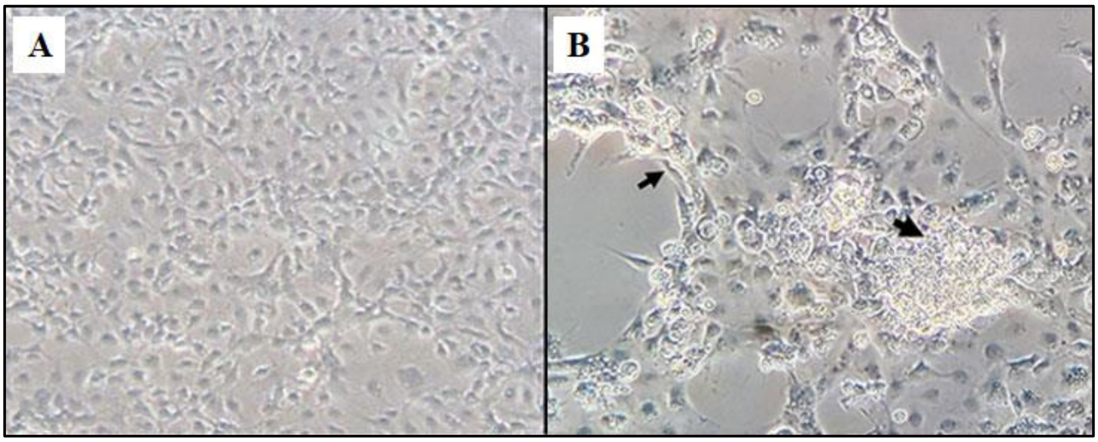

Figure 11. Cytopathic effect caused by TiLV in E-11 cell culture. (A) E-11 cells inoculated with tissue homogenate prepared from normal tilapia brain; (B) E-11 cells inoculated with tissue homogenate prepared from TiLV-positive brain, CPE with cell shrinkage and syncytial formation (black arrow), four days post-inoculation, magnification = 20 X. Source: Tattiyapong, Dachavichitlead and Surachetpong, 2017.

TiLV can also multiply in a primary brain cell line at 25 °C (Eyngor *et al*., 2014). But there is no plaque formation after the viral inoculation, the appearance of CPE was characterized by conversion of the typical elongated cells into swollen, rounded, and granulated cells at 10–20 days post-inoculation and leading to large monolayer detachment at days 14–19.

To preserve and store virus stocks, the TiLV-infected cell cultures can be centrifuged at 3 000 ×g for 15 minutes at 2–5 °C; the virus-containing supernatants can be dispensed into sterile vials at volumes of 0.3–0.5 ml each and stored at -80 °C or liquid N_2. For long-term storage, the virus can be stored through lyophilization process.

3.3.7 **Laboratory infection**

The laboratory infection protocols for testing the pathogenicity of TiLV in the healthy indicator fish have been described. The fish that were exposed to TiLV inoculum, exhibiting gross signs and mortality similar to TiLVD, need to be confirmed by histology and molecular tests (RT-PCR, RT-LAMP, ISH, immunodetection), or cell-culture assays.

(a) Challenge by intraperitoneal (IP) injection

For laboratory experimental infection, the healthy indicator tilapia (e.g. *O. niloticus*; weight: 30–35 g) should be maintained in a specific-pathogen-free (SPF) facility at 28 °C with aeration and fed with a commercial diet daily. The TiLV inoculum can be obtained from filtered (0.22 µm) tissue homogenate or from the cell culture supernatant (see above). Prior to IP injection, the healthy tilapia should be sedated using MS-222 (or other anaesthetic reagents) for 5 minutes and then injected intraperitoneally with 50 µL (depending on the size of fish) of TiLV-containing tissue homogenate or cell culture supernatant. An optimal dosage (e.g. 10^5 – 10^6 $TCID_{50}$) can be injected into each healthy fish. For the control group, fish can be injected with PBS or HBSS. The injected fish will be monitored for the appearance of gross signs and associated mortalities for two weeks. It has been shown that high mortalities in Nile (86%) and red tilapia (66%) occurred within 4–12 days post-injection (Tattiyapong, Dachavichitlead and

Surachetpong, 2017). Any dead or moribund fish should be removed and examined for the appearance of gross lesions, then the target tissues (e.g. liver, spleen, kidney, brain) can be sampled for TiLV RT-PCR (RT-qPCR), immuno-detection or histological evaluation.

(b) Challenge by cohabitation

Groups of tilapia will be kept in one aquarium divided into two (or more, depends on experiment design) compartments by water - permeable grids that allowed water circulation throughout the aquarium (Eyngor *et al*., 2014). The healthy indicator fish will be kept in one compartment, whereas TiLV-infected fish will be stocked into another compartment. This setup has been used to study the horizontal transmission of TiLV from infected tilapia (act as shedders) to naïve tilapia (Liamnimitr *et al*., 2018), and to test the susceptibility of other warm-water fish species (Jaemwimol *et al*., 2018).

(c) Challenge by intragastric gavage

To deliver TiLV orally, healthy tilapia will be anaesthetized, then gavaged with 0.1 ml of the TiLV - containing cell-culture supernatant at a dose of 1×10^4 $TCID_{50}$/fish with a 1 mL syringe connected to a flexible plastic animal feeding needle (20G × 1.5 inches long). The needle will be slowly inserted in the mouth and moved through the oesophagus until it reaches the stomach; the viral solution is then injected into the stomach. Using this procedure, Pierezan *et al*. (2019) have shown that challenged Nile tilapia (6 g) had a 40 percent mortality at 10 days after exposure; the gavaged fish exhibited gross signs of TiLVD. Histological examination of liver tissue also revealed syncytial cell formation, intracytoplasmic inclusion bodies and multifocal hepatocellular degeneration and necrosis. TiLV RNA was detected in faeces and gills of gavaged fish, as well as in the tank water holding the challenged fish. Further study revealed that the TiLV viral load are similar in gills and intestine with other systemic organs and demonstrate the intestinal routes as an important port of TiLV entry (Pierezan *et al*., 2020).

3.3.8 Corroborative diagnostic criteria

In a suspected TiLVD outbreak determined by the gross signs and/or the occurrence of mortalities, RT-PCR (RT-qPCR, or RT-LAMP) should be used as the initial confirmatory test as it provides a rapid turnaround. A definitive association can be made from the isolation of TiLV through cell-culture cultivation, or laboratory infection studies.

(a) Definition of suspect case

Infection of TiLV is suspected if at least one of the following criteria is met:

(1) gross signs and mortality consistent with TiLVD;
(2) histopathology consistent with TiLVD, e.g. formation of syncytial cells in the hepatocytes; and
(3) positive reaction by nucleic acid amplification methods, such as TiLV RT-PCR or RT-LAMP.

(b) Definition of confirmed case

Infection of TiLV is considered to be confirmed if two or more of the following criteria are met:

(1) TiLV virions can be isolated from tissues of infected fish, then confirmed by performing the cell-culture assays or laboratory infection using healthy indicator fish;
(2) histopathology consistent with TiLVD, e.g. formation of syncytial cells in the hepatocytes; and
(3) positive reaction from TiLV RT-PCR and amplicons' sequence analysis.

Although TiLVD is not an OIE notifiable disease, but countries with positive cases should report the information to OIE's World Animal Health Information System (WAHIS).

4. Prevention and control

4.1 *Fish immunity*

Fish immune responses are critical in the defence against pathogens. These immune responses consist of innate (non-specific) and adaptive (specific) systems.

The rapidly responding innate immunity is the first line of defence system against pathogens. Its two major components are phagocytic cells (neutrophils, macrophages, granulocytes), and humoral elements (such as antimicrobial peptides, lysozymes, lectins, complements, cytokines, etc.) (Secombes and Fletcher, 1992; Magnadóttir, 2006). The innate immune system is also involved in activating and regulating the adaptive immune system.

The adaptive immune system, which produces acquired immunity, is activated more slowly but develops a "memory" response that protects the host from attacks by pathogens to which it has been previously exposed. Activation of the adaptive immune system is relatively slow because it involves specific receptor selection, cellular proliferation, and protein synthesis. This adaptive immunity is long-lasting; its components involve cellular (T- and B-lymphocytes) and humoral elements (immunoglobulins). Three classes of immunoglobulins (antibodies), which are important in the establishment of adaptive immunity, are found in teleost fishes. These are IgM, IgT and IgD, of which IgM is the most prevalent.

4.2 *Vaccination*

Vaccination can be an effective prophylactic measure for controlling fish diseases. Vaccines are non-pathogenic, biological preparation to induce adaptive immunity against a specific pathogen. They are available for >17 species of fishes (including grass carp, salmon, trout, channel catfish, mandarin fish, tilapia) and protect against more than 22 different bacterial diseases (e.g. freshwater fish bacteria septicaemia, vibriosis, flavobacteriosis, columnaris, streptococcosis) and 8 viral diseases: infectious pancreatic necrosis (caused by IPNV), infectious salmon anaemia (ISAV), salmon pancreas disease (SPD), infectious haematopoietic necrosis (IHN), viral haemorrhagic septicaemia (VHS), red sea bream iridovirus disease (RSIVD), grass carp haemorrhagic disease (GCHD), infectious spleen and kidney necrosis (ISKN) (Brudeseth *et al*., 2013; Wang, Ji and Xua, 2020). The development and application of vaccines are relatively expensive, and so are employed mainly in large-scale, high-value, fish aquaculture (such as salmon and trout) in northern Europe, Chile, and North America.

The market for vaccines for tilapia is relatively small, in part because it is a low-value species. For tilapia, three vaccines have been produced and marketed by MSD (a trade name of Merck

& Co, New Jersey, USA). These are: 1) AQUAVAC® Strep sa (attenuated *Streptococcus agalactiae*) for streptococcosis, a disease caused by the infection of *S. agalactiae*;
2) AQUAVAC® Strep si (attenuated *S. iniae*); and 3) AQUAVAC® IridoV, for an iridovirus disease. Recently, tilapia culture has intensified, involving large-scale production; so interest in the development of vaccines for tilapia has increased. The development of TiLV vaccines are in progress in Israel, Thailand and China.

In a study of humoral response against TiLV, Tattiyapong *et al*. (2020) have shown that tilapia can produce anti-TiLV antibodies within 7–10 days after exposure to the virus. The antibody levels, in general, peaked at 2–4 weeks and then declined. When the fish were re-exposed (through IP injection) to TiLV, an increase in antibodies occurred within 2 weeks in most cases. This indicated that tilapia maintain humoral memory and produce anti-TiLV antibodies to protect fish from subsequent exposure. Recently, TiLV vaccine has been produced by β-propiolactone (BPL) inactivation (Zeng *et al*., 2021). This inactivated vaccine combined with the adjuvant Montanide IMS 1312 showed high efficacy, 85.7 percent relative percent survivals, in laboratory challenges. Specific IgM and neutralizing antibodies against TiLV were elicited at 3 weeks following primary immunization and were significantly increased after a booster. The mRNA levels of six immune-related genes: tumor necrosis factor-α (TNF-α), interleukin-1β (IL-1β), interferon γ (IFN-γ), cluster of differentiation 4 (CD4), and both major histocompatibility complex (MHC)-Ia and MHC-II, were all increased at 3–6 weeks after vaccination. The vaccine significantly lowered viral loads, indicating that TiLV-vaccine also inhibit viral proliferation. Most recently, Mai *et al*. (2021) also reported that both injectable heat-killed and formalin-killed vaccines are effective in preventing TiLV infection in Nile tilapia with relative percentage survival of 71.3% and 79.6%, respectively. The vaccines activated both branches of adaptive immunity, triggered expression of three immunoglobulin classes (IgM, IgD and IgT) and elicited not only systemic but also mucosal IgM responses in the vaccinated fish. These vaccines seem very promising for the prevention of TiLVD.

The vaccine dose and method of administration (e.g. injection, immersion, oral, spray) are usually based on the size of the fish. For large fish, vaccines can be administered through injection, which is the most effective in terms of level and duration of protection; but this delivery method is labour-intensive. Immersion and oral vaccines are used for smaller fish for ease of handling and the concomitant reduction in labour that is required; these vaccines are less expensive and not as stressful for young fish. In particular, oral administration is useful for fish that are cultured in large water bodies such as cages in the lakes. In some situations, vaccinations can be administered several times throughout the production cycle, for example, by starting with immersion vaccination for young fish, followed later with an oral/immersion vaccination as a booster, and finally with an injection for larger fish.

4.3 ***Immunostimulants***

Immunostimulants are substances that enhance the activity of one or more pathways of the immune response, acting to provide increased protection against pathogens. Immunostimulants act to enhance the general immune response and, unlike vaccines, do not target specific pathogens. Immunostimulants can supplement, or even be considered as an alternative to, vaccination protocols. Immunostimulants have broad-spectrum activity and are relatively cost-effective compared to vaccines. They are especially suitable for younger fish in which adaptive immunity is not fully developed.

There are various products on the market that contain β-glucans (long-chain polysaccharides), the most used immunostimulants, in fish aquaculture. These glucose polymers increase both the innate and adaptive immune pathways by enhancing phagocyte, humoral complement, and lysozyme activities. They also increase antibody responses (Sakai, 1999; Dong *et al*., 2015). β-glucans have been shown to be effective against both viral and bacterial diseases.

There has been increasing interest in the use of immunostimulants from plants with links to traditional medicine of Asia (Jeney *et al*., 2009). A number of these natural products have been developed and found to be effective in tilapia culture. For example, rosemary *Rosmarinus officinalis* was incorporated into diet to treat *Streptococcus* infection in tilapia (*Oreochromis* spp.) (Abutbul *et al*., 2004). It is shown that a diet supplemented with the Chinese herbal medicine plant *Sophora flavescens* significantly enhanced lysozyme production, cellular myeloperoxidase activity, antiprotease levels, and natural haemolytic complement activity in Nile tilapia. These fish fed *S. flavescens* had a mortality of 21 percent, compared to 80 percent mortality of the control group, when challenged with *Streptococcus agalactiae* (Wu *et al*., 2013). Studies of the effects of medicinal plants on expression of cellular immune effectors and resistance to pathogens in tilapia are reviewed by Kuebutornye and Abarike (2020).

4.4 ***Probiotics***

Probiotics are live microorganisms that provide health benefits to the fish. Many of these microorganisms, primarily bacteria and yeasts, are either found in normal pond environments or are components of the fish's intestinal microflora. Probiotics are either provided as feed additives, for strengthening immune responses, or dispersed in the pond to improve water quality (Cha *et al*., 2013). The use of probiotics against fish pathogens (bacteria and viruses) is a common practice in aquaculture. It has already been shown that probiotic bacteria can directly modulate fish immunity (both innate and adaptive) through elevating phagocytic activities, lysozyme levels, complement responses, and cytokine production (Salinas *et al*., 2005). A probiotic bacterium *Bacillus subtilis* C-3102 was demonstrated to enhance the adhesion of beneficial bacteria on gut mucosal surfaces of hybrid tilapia (*O. niloticus* × *O. aureus*) and to increase the expression of intestinal cytokines like IL-1b, TGF-β and TNF-α (He *et al*., 2013). Probiotic products selected for use should be those that are both effective and safe; they should not contain virulent or antimicrobial resistance genes (Kesarcodi-Watson *et al*., 2008). The benefit of probiotic supplementation (*Bacillus* spp.) was demonstrated to have a lower mortality (25%) in a TiLV challenge bioassay, a higher mortality (32%) was observed in the tilapia those were not fed probiotic diet (Waiyamitra *et al*., 2020). Tilapia fed with 1 percent probiotic diet had a significantly lower viral load than those fed with control diet. The transcription of immune-related genes, including *il-8*, *ifn-γ*, *irf-3*, *mx*, *rsad-2* (also known as VIPERIN) were significantly up regulated upon probiotic treatment.

4.5 ***TiLV-free tilapia***

To prevent TiLVD outbreaks, it is important to avoid the introduction of infected tilapia into aquaculture facilities. Most tilapia producers use SPF animals that are free of TiLV. The SPF tilapia are raised in biosecure facilities where their health status is monitored on an ongoing basis with sensitive, diagnostic methods. A declaration of freedom from TiLV should be based on monitoring for more than two consecutive years with at least two surveys (sampling and testing) per year conducted more than three months apart. The surveys should target the life stage that is most susceptible and be conducted at times of the year (such as summer) when TiLV is most likely to manifest to detectable levels (OIE, 2019a).

To be safe, any newly arrived tilapia stock should be quarantined for at least two weeks. Quarantine systems should be in a separate facility, away from existing populations, with dedicated equipment. During the quarantine period, the fish should be monitored for gross signs of TiLVD and mortalities, daily. Some farmers also include a stress test exposing the introduced fish at a high temperature (>25 ^{o}C) to ensure the robustness of the stocks.

4.6 *TiLVD-resistant tilapia*

One mechanism of disease resistance is that the virus is unable to enter the host cells due to the lack of virus receptors. These resistant fish are not susceptible to viral infection even when they are exposed to large quantities of viruses. A second type of resistance is viral tolerance in which either the virus is unable to replicate prolifically, or the virus is quickly eliminated by the host's immune system. The disease-tolerant host can still be infected by the virus but has a lower viral load and does not exhibit the clinical symptoms, nor the associated mortality.

There have been no reports describing the underlying mechanisms in TiLV-resistant lines of tilapia. Commercial providers use the term "resistant stock" without regard to the mechanism of resistance. Some of these stocks may be susceptible to viral infection and, therefore, capable of spreading the virus.

Stocking with resistant fish can be an effective management strategy to reduce the risk of production losses associated with the disease. Breeding programs selecting for disease resistance have been successful for a variety of fish species, including Atlantic salmon, rainbow trout, and tilapia. The success of such efforts depends on the heritability of the desired traits and the underlying genetic variability. With tilapia, several studies have estimated significant genetic variation for resistance to bacterial pathogens such as *Streptococcus iniae, Flavobacterium columnare* (LaFrentz *et al*., 2016; Wonmongkol *et al*., 2018).

Resistance to TiLVD varies among tilapia strains. Ferguson *et al*. (2014) found in Ecuador that a strain of all male tilapia (*O. niloticus* GMT) was more resistant to TiLV infection than the Chitralada strain (*O. niloticus*). Producers in Israel found that resistance to TiLVD was highest in red tilapia, followed by the Chitralada strain with hybrid tilapia being the least resistant. But in Thailand, Nile tilapia are more resistant (cumulative mortalities of <30% from TiLVD outbreaks) than red tilapia (cumulative mortalities of 30–100%) (Tattiyapong, Dachavichitlead and Surachetpong, 2017). Attributing variation of TiLVD resistance to genetics is complicated because of the potential effects from coinfecting bacteria, e.g. *Streptococcus* spp. and *Aeromonas* spp. (Amal *et al*., 2018; Nicholson *et al*., 2020) as well as other viruses (Yamkasem *et al*., 2021c). The impact of TiLV-*Aeromonas hydrophila* coinfection was demonstrated in laboratory challenge studies, where coinfected tilapia had a 93 percent cumulative mortality as compared to 34 percent, and 7 percent in the TiLV- or *A. hydrophila*-infected groups, respectively (Nicholson *et al*., 2020).

Based on a natural outbreak in a pond stocked with mixed families of genetically improved farmed tilapia (GIFT), Barría *et al*. (2020) used statistical analysis of survival, relative time to mortality, and family lines to conclude that resistance to TiLVD is highly heritable and does not diminish growth. Thus, it seems that selective breeding to increase TiLVD resistance may be feasible.

Furthermore, Barría *et al*. (2021) conducted a genome-wide association study (GWAS) to explore the genetic component of TiLVD resistance. The study involved 950 GIFT tilapia, categorized as either survivor (resistant) or mortality (susceptible), from a TiLV outbreak. These fish were genotyped with a 65 K single-nucleotide polymorphism (SNP) array. A quantitative trait locus (QTL) of large effect, associated with resistance, was identified on chromosome *Oni22*. The average mortality rate of tilapia that were homozygous for the resistance allele at the most significant SNP was 11%, compared to 43% for fish that were homozygous for the susceptibility allele. Several potential candidate genes on *Oni22* were identified, including *lgals17*, *vps52*, and *trim29*, genes that have been linked to processes involved in host immunity and viral replication. The discovery of genetic markers linked to resistance can be used to improve selective breeding and could serve as a pathway to future gene editing protocols to reduce production losses from TiLVD.

5. Epidemiology

5.1 *Geographic distribution*

TiLVD has, so far, been reported in the following 16 countries and provinces:

In **Israel**, TiLV was first officially detected through diagnosis in samples collected in 2011 (Eyngor *et al*., 2014; Bacharach *et al*., 2016). However, TiLVD is suspected of having been responsible for earlier fish mass mortalities. For example, in the Sea of Galilee in 2009, the fish catch declined to 7 tonnes, down from 287 tonnes in 2005. Of the ~27 fish species (families Cichlidae, Cyprinidae, Mugillidae, and Claridae) inhabiting the lake, only the tilapias declined substantially. Also, during this time, incidences of mortality of tilapia began to occur in commercial farms located on the northern coastal shore, Bet-Shean, Yizrael, the Jordan Valley, and Upper and Lower Galilee. Die-offs occurred primarily during the hot season (May-October). After the initial wave of mortality in a pond, the disease outbreak ended, suggesting that the surviving fish have immunity against TiLV. In polyculture farms, only tilapia were affected; cohabiting carp *Cyprinus carpio* and grey mullet *Mugil cephalus* were not.

An investigation of tilapia mortality events on 14 Israeli fish farms was conducted during 2017–2018. Of the 89 samples (3–4 fish per sample, lethargic individuals with or without obvious clinical signs) analysed, 42 percent were TiLV positive (Skornik *et al.*, 2020). The sequences of the genomic segment 3 of 25 of these TiLV isolates were compared with those of other TiLV isolates published in GenBank. Phylogenetic analysis revealed three distinct clades: clade-A, including 50 isolates from Israel (2011, 2017–2018), Ecuador (2012), Egypt (2015), Thailand (2015–2019), India (2017), Peru (2018), USA (2018–2019); clade-B, consisting of 4 isolates from Thailand (2018) and Bangladesh (2017 and 2019); clade-C, containing a single Bangladesh isolate (2019). Within clade-A, 14 Israeli 2018 isolates formed a distinctive sub-clade (IL-2018), all from lower (31.2±25.6%) survival events. The average survival was 58.1±21.5 percent in the overall mortality events within clade A. This suggests that TiLV may have become more virulent during 2018.

In **Ecuador**, mass die-offs of tilapia were reported since 2009, presumably resulted from TiLVD outbreaks (Bacharach *et al*., 2016). This disease was first officially confirmed in a case from a 2012 outbreak that resulted in a 99 percent mortality, where affected fish were found to have lesions in the liver consisting of necrotic hepatocytes and syncytial cell formations (Ferguson *et al*., 2014; Tsofack *et al*., 2017). From samples collected during this outbreak, TiLV strain EC-2012 was isolated and sequenced (Subramaniam *et al*., 2019). From samples of a previous outbreak (unknown year, prior to 2012) TiLV was isolated, sequenced, and compared by Bacharach *et al* (2016) to the Israel strain TiL-4-2011. They showed that these two strains are highly similar, with 97–99 percent nucleotide sequence identity.

In **Colombia**, liver samples from farmed tilapia with syncytial hepatitis were analysed by RT-PCR and found to be positive for TiLV (Tsofack *et al*., 2017). The year that the samples were collected, and the location of the affected farm is not known. In 2016, TiLV was detected in the department of Córdoba, area of the Colombian Caribbean (Contreras *et al*., 2021). Of the seven farms surveyed, three (42.85%) were positive for TiLV. These TiLV-positive fish exhibited clinical signs typically seen in infected fish, such as eroded fins, eye lesions, abnormal behavior (lethargy, anorexia). By RT-PCR, eighteen (27.27%) were infected with TiLV from 66 fish sampled. The virus was detected in the brain (64.3%, 18/28), spleen (61.9%, 13/21), and liver (35.7%, 10/28).

In **Thailand**, TiLV outbreaks in Nile tilapia and red tilapia (*Oreochromis* spp.) occurred in 2015 (OIE 2017a; Surachetpong, 2017; Surachetpong *et al*., 2017). The outbreaks occurred at fish farms in central, western, eastern, and northeastern Thailand. In general, clinical signs and high mortality rates (20–90%) were found in smaller fish (1–50 g). TiLV is suspected to be responsible for the previously observed tilapia one-month mortality syndrome (TOMMS), in which mass mortality occurred one month after transfer of fish from hatchery to grow-out cages.

In **Egypt**, Fathi *et al*. (2017) reported that during the early 2010s, some producers found that populations of medium (>100 g) and larger sized Nile tilapia often experienced high mortalities during the summer months (June-October, water temperature 25 ^{o}C); phenomenon known as summer mortality syndrome (SMS), with yet unknown causative agent. In an epidemiological survey conducted in 2015–2016, in the regions of Kafr El Sheikh, Behera and Sharkia, Fathi *et al*. (2017) reported that 37 percent of fish farms were affected by SMS, with average mortalities of 9.2 percent on affected farms. Fish tissues (brain, kidney, liver and spleen) samples from affected farms tested TiLV-positive by RT-PCR. The amplicon sequence from the samples from Egypt was 93% identical to isolates from Israel. Another investigation in the Nile delta region showed that, in 2015, four out of eight tilapia farms (50%) were positive of TiLV by RT-PCR and verified with amplicons' sequencing (Nicholson *et al*., 2017).

In the **United Republic of Tanzania**, TiLV was detected by RT-PCR in samples of farmed and wild Nile tilapia collected from Lake Victoria in 2015 (Mugimba *et al*., 2018). Among the 108 wild fish sampled, at Maganga beach and Mchongomani, 18 were positive, a prevalence of 16 percent. TiLV was detected primarily in tissues of the spleen and kidney. It was found less frequently in the liver and was not found in the brain.

In **Uganda**, TiLV was detected by RT-PCR in wild and cage-farmed tilapia in Lake Victoria in 2016 (Mugimba *et al*., 2018). Among 83 fish sampled, 10 fish were positive for TiLV, a prevalence of 12 percent. TiLV was detected in spleen, kidney, and sometimes liver, but not in brain tissue.

In **India,** TiLVD outbreaks occurred in farmed tilapia in the eastern state of West Bengal and southwestern state of Kerala (Behera *et al*., 2018). Diseased fish exhibited clinical signs of TiLVD with >85 percent mortality, and the presence of TiLV was confirmed by RT-PCR. None of the cohabiting fishes: Indian major carps (*Labeo rohita*, *Catla catla*, *Cirrhinus mrigala*), milkfish (*Chanos chanos*) and pearl spot cichlid (*Etroplus suratensis*) were affected. During

November 2018, mass mortality was observed in wild tilapines in a lake in Katpadi, Vellore, the affected fish showed red skin, scale protrusion, swollen eyeballs and swam erratically. Through RT-PCR and histological examination, TiLV was detected (Saranya and Sudhakaran, 2020).

In **Indonesia**, TiLV was detected by RT-PCR in tilapia cultured in islands of Java, Lombok, and Sumatra during 2016; the clinical signs included exophthalmia, corneal opaque and/or cataract, as well as skin abrasion and darker body coloration. The mortalities reached 70–100 percent (Koesharyani *et al.*, 2018) in some ponds.

In **Taiwan Province of China,** TiLV was detected by RT-PCR in blue hybrid tilapia farmed in Guanyin district, Taoyuan City in 2017. There, 9 600 out of 150 000 fish died, a mortality of 6.4 percent (OIE, 2017b).

In **the Philippines**, TiLV was detected in 2017 by RT-PCR assays. TiLVD-associated mortality was observed in a nursery pond, the mortality (~25%) appeared 15 days after stocking the pond (OIE, 2017c).

In **Malaysia**, TiLV was detected, by both RT-PCR and histology examination, in farmed red hybrid tilapia in 2017 (OIE, 2017d; Amal *et al*., 2018). Mortalities began 45 days after stocking the fry into grow-out ponds and continued for three weeks. The mortality rate peaked 5–9 days after the first noticeable die-off. The cumulative mortality was 25 percent.

In **Bangladesh**, TiLV was detected in 2017 in populations of farmed Nile tilapia in Tishal Upazila, Mymensingh District (Chaput *et al*., 2020). The virus did not infect co-cultured fishes: pangasius catfish, or carps (rohu and common carp). Hossain *et al.* (2020) reported the presence of TiLV in 9 farms from 36 farms surveyed, the outbreaks occurred among fingerlings within 10–15 days post-transfer to grow-out ponds, with mortality ranging from 15–20 percent to 82 percent during 2016–2018. Debnath *et al* (2020) also performed TiLV surveillance in tilapia farms and hatcheries in 2017 and 2019. Among farms experiencing unusual mortality, 8 out of 11 farms tested positive for TiLV in 2017, and 2 out of 7 tested positive in 2019. Investigation of asymptomatic broodstock collected from 16 tilapia hatcheries revealed that 6 hatcheries tested positive for TiLV.

In **Peru,** TiLVD outbreaks first occurred in late 2017 in the Piura region, in wild populations of Nile tilapia from the Poechos Reservoir and on nearby farms (OIE, 2018a). During January to April 2018, the virus spread from the Piura region to farms in San Martín, the region with the most tilapia farms, and to the Lambayeque region. Later, in Sept 2018, TiLV spread from San Martin into farms located in the Cajamarca region. The following year (2019), the virus was introduced from San Martin to the Huanuco region. The mortalities associated with TiLVD ranged from 20 percent to 100 percent on commercial farms. The nucleotide sequence of the TiLV isolate from San Martin (collected February 2018) was compared to the sequences of isolates from Israel and Thailand by Pulido *et al*. (2019). They found 97 percent and 95 percent identities to the Israeli and Thai isolates, respectively; and confirmed, through multilocus sequence phylogenetic analysis, that the Peruvian isolate is closely related to the Israeli isolate.

In **Mexico,** TiLV was detected by RT-PCR in 20 tilapia farms in July 2018 by an active surveillance program conducted in farms in six states: Chiapas, Jalisco, Michoacán, Sinaloa, Tabasco, and Veracruz (OIE, 2018b).

In the **United States of America (USA),** TiLV was detected in the states of Idaho, Wyoming, and Colorado in 2018–2019 (OIE, 2019b). One Idaho farm had a history of importing live tilapia from Thailand. Two isolates of TiLV obtained from this farm were sequenced and subjected to phylogenetic analysis. As expected, both isolates were shown to be most closely related to Thai strains of TiLV (Ahasan *et al*., 2020).

The country and the first year that TiLVD was detected are shown in Table 4.

Table 4. Geographic distribution of tilapia lake virus disease.

Country/province	**Year TiLVD detected /first peer-reviewed publication**	**OIE notification***
Israel	2011/Eyngor *et al*., 2014	2017
Ecuador	2012/Ferguson *et al*., 2014	
Colombia	unknown/Tsofack *et al*., 2017	
Thailand	2015/ Surachetpong *et al*., 2017	2017
Egypt	2015/ Fathi *et al*., 2017	
United Republic of Tanzania	2015/ Mugimba *et al*., 2018	
Uganda	2016/ Mugimba *et al*., 2018	
India	2016/ Behera *et al*., 2018	2017
Indonesia	2016/ Koesharyani *et al*., 2018	
Taiwan Province of China	2017	2017
The Philippines	2017	2017
Malaysia	2017/ Amal *et al*., 2018	2017
Peru	2017/ Pulido *et al*., 2019	2018
Mexico	2018	2018
United States of America	2019/ Ahasan *et al*. 2020	2019

*TiLVD was determined to be an emerging disease by OIE in 2016, and member countries began reporting the disease through WAHIS.

5.2 *Genotype*

Phylogenetic analyses based on the whole TiLV genome sequence (concatenating ORF-coding regions of 10 genomic segments), reveals two distinct genotypes: 1) Israel genotype consisting of isolates from Israel, Ecuador and Peru; 2) Thai genotype clustering isolates from Thailand, Bangladesh and US (Pulido *et al*., 2019; Chaput *et al*., 2020; Debnath *et al*., 2020).

5.3 *Persistence in the environment*

It is important to know how long TiLV can survive in the environment to determine the duration of holding fish-free intake water in a reservoir. The virus can be shed into pondwater and infect other tilapia. However, no data are available regarding persistence of the TiLV virions in pondwater, pond sediments, or on farm equipment. But TiLV is expected to possess similar properties to orthomyxo-like viruses found in fish, such as ISAV, which remains infectious after 9–18 days in filtered estuarine water and seawater at 28±2 °C. It was not known if there is a difference in stability of orthomyxo-like virus between seawater and freshwater. Falk *et al*. (1997) showed that ISAV were stable in the pH range 5–9. At pH 4 the virus was completely inactivated after 30 minutes, at pH 11, a 90 percent reduction in infectivity was observed after 30 minutes. Incubation at 56 °C completely inactivated the virus in 5 minutes.

5.4 *Reservoir host*

The potential reservoirs are sub-clinically infected farmed or wild tilapia and giant gourami.

5.5 *Modes of transmission*

5.5.1 Vertical transmission

Recent studies suggest TiLV can be transmitted vertically. Tissues of reproductive organs, ovary and testis, collected from experimentally infected broodstock tested positive for TiLV by RT-PCR, histological evaluation, and cell-culture system; fertilized eggs produced by infected fish also tested positive by RT-PCR and ISH (Dong *et al*., 2020). Yamkasem *et al*. (2019) also detected the presence of TiLV genomic RNA and viable viruses in the ovary and testis of naturally and experimentally infected tilapia, as well as 2-day-old fry from infected broodstock.

5.5.2 Horizontal transmission

TiLV can be transmitted horizontally. From cohabitation bioassays, it is clear that susceptible individuals that cultured with TiLV-infected fish in the same tank can become infected and results in a high mortality (56–80%) within 8–10 days (Eyngor *et al*., 2014; Liamnimitr *et al*., 2018). Horizontal transmission is likely the main route for the spread of TiLVD; the transmission could occur through cannibalism of infected sick/dead fish, or from the mucus of moribund/dead fish.

5.6 *Risk factors*

Environmental conditions affect both TiLV and the fish host which are important determinants of disease outbreaks. Environmental factors affect TiLV multiplication and play a relevant role in the clinical manifestation of disease. The host is affected by conditions such as temperature, water quality, stocking density, stress (from handling and transport); fish are more susceptible to disease when stressed from sub-optimal culture conditions. Tilapia are tolerant to a wide range of environmental conditions, e.g. high salinity, the Nile tilapia can grow at salinity up to 25 ppt (parts per thousand), blue hybrid tilapia at 20 ppt, and red tilapia can tolerate salinity up to 32 ppt. Tilapia are less tolerant to low water temperature, and generally stop feeding when water temperature falls below 18 ^{o}C. Today, more than 70 percent of commercial tilapia is Nile tilapia, its optimal water quality parameters are listed in Table 5.

Table 5. The optimal temperature and water quality for Nile tilapia.

Growth condition	Optimum	Range
Temperature (^{o}C)	27–30	12–38
Salinity (ppt)	5–10	<25
Dissolved oxygen (mg/L)	>5	
pH	6–9	5–10
Ammonia (NH_3) (mg/L)	<0.1	
Nitrate (NO_3^-) (mg/L)	<27	

5.7 *Risk assessment*

The FAO conducted a TiLVD risk assessment survey through interviews of experts, a technique known as “expert knowledge elicitation” (EKE) (FAO, 2018b). The major findings indicate that:

(a) TiLVD represents a significant risk to those parts of the world where tilapia aquaculture or fisheries is important to either food security or commerce, especially in Asia, Africa and South America;
(b) the major risk of spreading the disease is through the translocation of live fish;
(c) uncooked, chilled/frozen whole fish and fish products (such as fillets) has a lower risk for disease spread; and
(d) potential control measures involve restrictions on the transport of live fish, frequent surveillance of stocks, biosecurity, and the development of emergency response plans.

TiLV can be inactivated by prolonged freezing. Thammatorn, Rawiwan and Surachetpong (2019) found that TiLV was inactivated in tilapia fillets stored at -20 ^{o}C for 3 to 4 months, suggesting that properly frozen products (e.g. whole fish or fillets) are not a risk for spreading the virus. This result is consistent to that of EKE reported above.

5.8 *Impact of the TiLVD*

Outbreaks of TiLVD result in substantial losses to producers of farmed tilapia. The reduction in fish production may subsequently have negative impacts on related, or dependent, industries such as fish feed manufacturers; processors; brokers; and exporters. Socioeconomic impacts of TiLVD could also be felt in some areas. For example, in tilapia farming areas of Africa, many small-scale producers sell directly to buyers, often women, whose livelihoods depend on re-selling the fish in their communities. In such cases, tilapia production losses can have effects beyond the farm gate.

Potentially, tilapia fishery stocks can be negatively impacted. Eyngor *et al*. (2014) found that the wild populations of *Sarotherodon galilaeus* in the Sea of Galilee were infected with TiLV, although this was not associated with the mass mortalities observed in infected populations of farmed fish in the region.

6. Principles of eradication, containment, and mitigation

This section provides information to aid the development of strategies for responding to TiLVD outbreaks in farmed tilapia populations. The major goals in responding to TiLVD outbreaks are: 1) to eradicate the disease where possible; 2) to prevent its spread; and/or 3) to reduce the impact of the disease. The specific methods employed in achieving these goals will depend on a variety of factors including the extent of the outbreak and the type of production facilities involved (e.g. recirculation systems, ponds, raceways, cage-culture, etc.).

- *Eradication*. Eradication of TiLVD from a country will be possible if TiLV has not spread to wild fish populations. Eradication is most likely to be successful if the outbreak has been rapidly detected and is localized.

- *Containment*. In areas where TiLV has entered natural waters and become established in wild populations, country-wide eradication may not be possible. Measures can be taken, however, to prevent further spread of the disease through containment to affected premises (areas) where control efforts are continued.
- *Mitigation*. This measure involves the implementation of management practices that reduce the incidence and severity of TiLVD outbreaks.

6.1 *Eradication*

Eradication of this disease is feasible for production facilities that are under strict management. If possible, the origin of the TiLV introduction should be determined to prevent its re-entry into disinfected areas.

Eradication of TiLVD from an infected production facility will involve the following steps:

(a) **Diagnosis** – Obtaining an accurate diagnosis is the first step in disease management. From moribund/live fish with gross signs of the disease, samples are sent to a competent diagnostic laboratory. Upon confirmation of the diagnosis, the prevalence of the disease in the farmed populations can be determined through surveillance sampling.

(b) **Destruction and disposal of diseased stock and disinfection** – Diseased stocks must be removed. Disposal of wastes should be carried out immediately in a manner that prevents access by birds or other animals. Usually, such wastes are disposed of by burial. The facility should then be thoroughly disinfected. If possible, the water in the facility should be emptied and not re-stocked for a period of time. The period of fallowing depends on the types of culture system (earthen pond, tank, or raceway), weather conditions (dry, or raining season). Without the hosts/vectors, TiLV will then begin to degrade. The fallowing period is relevant to how long the virus can retain infectivity outside the hosts, a three-week period should be sufficient (see section 5.3 persistence in the environment).

(c) **Restocking with SPF/SPR stocks** – Once the facility is disinfected, it can be re-stocked with stocks certified to be SPF (TiLV-free) and/or SPR (TiLVD-resistant). It is important to obtain SPF/SPR stocks from reliable sources.

(d) **Management of facilities restored to TiLVD-free status**
Farms that have been restored to TiLVD-free status will have to implement strict management protocols in order to maintain this status. This will entail stocking the facility with SPF fish and ensuring high levels of biosecurity. The biosecurity measures detailed below are applicable to land-based operation system (pond, tank, and raceway), but many do not apply to cage-culture systems.

- establishing physical barriers, such as fences, to prevent unauthorized access to production facilities;
- restricting visitor activities (e.g. requiring sign in/out, disinfecting footwear, providing hand wash stations, etc.);
- refusing entry to individuals who have recently (within the past three days) been at a TiLVD-affected facility;
- disinfecting filtered intake water, such as with ultraviolet (UV) light or ozone;

- equipping production facilities with water reservoirs for water exchange;
- avoiding transfer of non-sanitized equipment (i.e. scoop nets, buckets, sampling containers, etc.) among ponds or tanks;
- cleaning, disinfecting and drying the rearing units between production cycles. Facilities should be cleaned with brushes and water jets and wastes diverted into a sanitary sewer, a sedimentation pond, or other effective treatment system. Several disinfectants including 2.5 ppm iodine, 10 ppm sodium hypochlorite (NaOCl), 300 ppm hydrogen peroxide (H_2O_2), 80 ppm formalin, and 5 000 ppm (0.5%) Virkon™ have been shown to be effective at inhibiting TiLV replication (Jaemwimol *et al*., 2019). In addition, the infectivity of TiLV is destroyed after 30 minutes of contact with commonly used disinfectants: buffered povidone iodine (PVP iodine; ≥50 ppm) complex and household bleach (free chlorine; ≥20 ppm) (Soto, Yun and Surachetpong, 2019);
- using TiLV-free or TiLVD-resistant stocks, obtained from reliable sources;
- introducing stocks with a known health history and quarantining the fish if needed. The general guidelines are described by Arthur, Bondad-Reantaso and Subasinghe (2008);
- performing routine health inspections of all stocks. Samples should be taken in accordance with approved protocols and tested for the presence of TiLV;
- keeping accurate records of mortalities and water quality;
- maintaining records of the source and movement of stocks; and
- implementing good feeding regimes with adequate nutrition, and prevention of hazardous contamination.

For fish cultured in cages in the open waters (lakes, rivers, reservoir, etc.), most of the biosecurity components described above would either not be applicable or would be impossible to implement. For cage-culture, the recommendations include:

- disposing of the sick/dead fish on land and not in the river. Disease can spread through the water to the farms downstream. The diseased fish should be promptly removed and buried, away from the open water;
- cleaning the nets, cages should be cleaned on land with brushes and water jets;
- limiting fish stress through good husbandry practices;
- stocking healthy fish or TiLVD-resistant stocks if available (see section 4.6 TiLVD-resistant tilapia); and
- administering vaccines once they become available and cost-effective. TiLVD vaccines are under development in Israel, China and Thailand.

6.2 *Containment*

If eradication of TILVD is not feasible following an outbreak, zoning and associated disease control measures should be implemented to prevent the spread of the virus to TiLV-free areas.

6.2.1 Quarantine and movement controls

Quarantine and movement restrictions should be implemented immediately upon suspicion of TiLVD outbreak. The CA should establish appropriate zone and compartment designations. Zoning and compartmentalization are management strategies to limit the spread of disease and facilitate international trade in fish and fish products. Zoning is usually under the responsibility of the CA; while compartments are within the production facilities and are managed through farm-level biosecurity programmes to maintain their health status. Detailed information regarding zoning and compartmentation can be found in the OIE and FAO publications (OIE, 2019a; Subasinghe, McGladdery and Hill, 2004; Tang, Bondad-Reantaso and Arthur, 2019). Basically, there are three premises/zone:

- *Affected premises (or area)*: the premises (e.g. farm) or area (e.g. geographically separated area) where TiLVD outbreak occurs, and its immediate vicinity. An affected premise or area is where a presumptive or confirmed positive case exists based on diagnostics results.
- *Buffer zone*: an area adjacent to the affected premises (or area); it requires stringent surveillance to prevent disease spread to the free zone.
- *Free zone*: the non-affected area.

A control area consists of the affected premises (or area) and the buffer zone.

Movement controls from the affected premises (or area) should include:

- bans on the movement of live, fresh (chilled on ice) fish from the affected premises into TiLV-free areas;
- bans or restrictions on releasing live fish and pondwater from the affected premises into aquatic environments;
- restrictions on discharging of processing plant effluent within the affected premises;
- restrictions on harvesting and then transporting TiLV-infected fish in the affected premises to off-site processing plants;
- restrictions on the use and movement of equipment and vehicles between farms within the affected premises;
- control of bird access to live and moribund/dead fish within the affected premises; and
- control of the disposal of diseased fish

The implementing of these bans or restrictions will depend on the severity of the disease, the types of operation (such as farm location, farm size, culture system), and the response options chosen.

6.2.2 **Tracing**

Tracing refers to investigation of: (a) if TiLVD has spread to other areas, (b) the origin(s) of the disease. Movements of the following from affected premises might need to be traced:

- live fish: for example, fry, fingerlings and broodstock and other stocks;
- fresh fish: intended for human consumption;
- effluent and waste products from processing plants and farms: discharge into nearby water bodies;
- vehicles: potentially contaminated transport vehicles, feed trucks, cars and boats; and
- farm materials: nets, buckets and other farm equipment

6.2.3 **Surveillance**

Surveillance is an ongoing systematic collection, analysis, interpretation of disease data through sampling of fish populations and monitoring the presence of TiLV. It can be used to detect the early occurrence of TiLVD and determine its prevalence in populations and is used in the process of maintaining and certifying farms (or areas) as being TiLV-free. Detailed information on general requirements for surveillance to establish freedom from TiLV at various prevalence thresholds is provided in the OIE Aquatic code (OIE, 2019a). FAO has implemented a TiLVD active surveillance programme designated for non-specialists using a tool kit containing 12-point checklist (Bondad-Reantaso *et al*., 2021).

6.3 ***Mitigation***

If the disease cannot be eradicated from natural waters or effectively contained to specific areas, producers may still operate successfully if farms are managed to keep TiLVD at a lower prevalence and severity. These management protocols should be aimed at reducing other forms of stress to farmed stocks such as by avoiding overcrowding, maintaining optimal environmental conditions, ensuring good water quality and the use of SPF/SPR stocks. Farms can continue to produce tilapia, but they may have to prohibit exports of live or fresh TiLV-infected fish to areas that are TiLV-free. Effective mitigating measures geared at curbing TiLVD outbreaks in fish farms include strict adherence to biosecurity and good aquaculture practices.

In addition, outbreaks of TiLVD may be triggered when the farmed populations are under increased stress, which may come from a variety of sources such as high stocking density, stress from transporting and handling, poor water quality or other less optimal environmental conditions (e.g. high temperature). To ameliorate these problems, farmers should attempt to maintain good pond conditions through installing sunscreens to lower the temperature, increasing aeration and water exchange, carefully monitoring, adjusting the feeding regime, applying probiotics, and removing the uneaten feeds.

Two main factors form the basis of reducing the impact of TiLVD outbreaks through pond management. TiLVD outbreaks primarily occur during warmer months and affect mainly smaller fish. Thus, farmers reduce the risk of production losses by not stocking or culturing susceptible tilapia (300 g or less) during the summer months. For example, in some countries, when a farmer starts production in early November by stocking fingerling (~20 g), the harvest size of approximately 500 g can be reached prior to the start of the summer (June through October). Other examples, farmers stock 2–3 times more fingerlings to compensate the losses during the disease outbreak. TiLVD-related production losses can be minimized through such cost-effective management practices.

6.4 ***Vector control***

There have been no reports of specific vectors of TiLV; however, piscivorous birds have been shown to be mechanical vectors, through their faeces or regurgitated food, for other fish viruses. Herons, for example, were found to be vectors for infectious pancreatic necrosis virus (IPNV), viral haemorrhagic septicaemia virus (VHSV) and spring viraemia of carp (SVCV) (Peters and Neukirch, 1986). IPNV is a stable, non-enveloped, virus that can survive the acidic conditions and the actions of digestive enzymes in the bird gut; and IPNV present in bird faeces remains infectious (McAllister and Owens, 1992). Both VHSV and SVCV are enveloped viruses, containing lipid membranes; these viruses are rapidly inactivated by the gastric lipases and proteases in the stomachs of birds but can be found in regurgitated samples. TiLV is also an enveloped virus and thus, vulnerable to the digestive enzymes and acidic conditions of the bird stomach. However, TiLV could, potentially, be spread by birds regurgitating the remains of infected fish into disease-free fish ponds. Piscivorous birds can often be found in large numbers in fish ponds throughout the year. The risk of spread of TiLV by piscivorous birds can be reduced by installing bird deterrents (e.g. electric fence, pond netting) around the ponds and by ensuring that they do not have access to disposed fish.

6.5 *Environmental considerations*

Care should be taken to ensure that TiLV is not introduced into wild populations of, either native or established, tilapia. Wild populations of tilapia are important components, in many areas, of commercial and artisanal fisheries. The spread of TiLV into these populations could decimate these fisheries and negatively impact the communities that are dependent on them. In addition, if TiLV spreads to wild populations of tilapia, it would be impossible to eradicate TiLVD from the area.

For all but cage culture systems, a management plan should be developed to prevent the escape of infected tilapia into natural waters where tilapia populations are present. If TiLV is detected, the infected sick/dead fish should be removed promptly and disposed of according to an established guide, such as the OIE Aquatic code (OIE, 2019a); they should not be disposed of into open waters (e.g. lakes, rivers). For cage culture in water bodies with wild tilapia populations, extra care should be taken to prevent bringing in diseased stocks because of the high probability of TiLV in caged tilapia populations being spread to wild ones.

6.6 *Public awareness*

To reduce the potential spread of TiLV, information regarding the disease and its management must be made available to producers and to relevant government agencies and technical staff. TiLVD is included as an emerging disease that should be reported to the OIE. Industry professionals should be aware that live fish imports from diseased areas (or countries) requires appropriate control measures (e.g. diagnostic certification, self-quarantine at farm level) to avoid disease introduction and spreading. Producers need timely information on TiLVD outbreaks to assess their need to implement or to enhance their biosecurity protocols to prevent introducing disease to their facilities.

Information regarding the TiLVD outbreaks status, and actions being taken to eradicate and control the disease, can be disseminated in a variety of forms such as brochures, agency technical reports, industry bulletins, and scientific publications. Information can also be distributed verbally through farm visits, aquaculture workshops, seminars, and symposia.

6.7 *Trade and industry considerations*

TiLV is primarily spread through international and domestic trade of live fish. Tilapia producers purchase “seed stock” (fry), or fingerlings from specialized commercial hatcheries or aquaculture centres. Large hatcheries may ship fry to many countries and, if their stocks become infected, they have the potential of spreading the disease around the globe as suggested by Dong *et al* (2017b). For example, the TiLV detected in a farm in Idaho, United States of America, is originated from a Thai hatchery as reported by Ahasan *et al*. (2020); this farm had history of importing live fish from Thailand. Regional hatcheries, similarly, have the potential to spread TiLV within a country or among neighbouring countries. For example, in a Peruvian case, the virus spread from San Martín (the province that has the highest concentration of tilapia farms) to the grow-out farms located in regions of Lambayeque, Cajamarca, and Huanuco (OIE, 2018a). To protect against the spread of TiLV, government agencies could institute a certification program to ensure that all imported stocks are free of TiLV. For example, after TiLVD outbreaks, the United States of America's Department of Agriculture has issued a Federal Order that requires imported shipments of live fish, fertilized eggs and gametes from TiLV-susceptible species to have an official health certificate and veterinary inspection from the exporting countries to certify TiLV-free status. Similar certification for TiLV-free status in regional or local markets may also be desirable. For trading commodity products, such as frozen, processed (dried, cured, smoked), cooked, or value-added (such as breaded, battered) tilapia, the safety of fish products is addressed in the OIE Aquatic code (OIE 2019a).

7. Policy development and implementation

7.1 *Overall policy*

Addressing biosecurity requires significant resources, strong political will and concerted international action and cooperation. National strategic planning for aquatic animal health and biosecurity is vital; without it, a country can only react in a piecemeal fashion to new developments in international trade and serious transboundary aquatic animal diseases, and its aquaculture and fisheries sectors will remain vulnerable to new and emerging diseases. The FAO encourages member countries to develop and formalize National Strategies for Aquatic Animal Health (NSAAH) and health management procedures (FAO, 2007).

A NSAAH is a broad yet comprehensive strategy to build and enhance capacity for the management of national aquatic biosecurity and aquatic animal health. It contains the national action plans at the short-, medium- and long-terms using phased implementation based on national needs and priorities. The NSAAH outlines the programmes and projects that will assist in developing a national approach to overall management of aquatic animal health and includes an Implementation Plan that identifies the activities that must be accomplished by government, academia and the private sector. The draft framework of the NSAAH should be discussed with and accepted by key stakeholders via a public-private partnership. The final document should be distributed to national policymakers, aquaculturists, other private stakeholders and the public; and the NSAAH should be endorsed by the Competent Authority as an official policy document.

The development of a NSAAH includes a gap analysis (achieved through a self-assessment survey and strengths weaknesses opportunities and threats analysis) conducted by national and regional focal points, a committee or a task force, a working group on aquatic animal health or any structure that fits the country. The technical elements that may be considered in the strategic framework will vary depending on an individual country's situation, and thus may not include all the programme elements listed below (alternatively, additional programmes may be identified as having national and/or regional importance and thus need to be included):

i. Policy, Legislation and Enforcement
ii. Risk Analysis
iii. National Aquatic Pathogen List
iv. Health Certification, Border Inspection and Quarantine
v. Disease Diagnostics
vi. Farm-level Biosecurity and Health Management
vii. Use of Veterinary Drugs and Avoidance of AMR
viii. Surveillance, Monitoring and Reporting

ix. Communication and Information Systems
x. Zoning and Compartmentalization
xi. Emergency Preparedness and Contingency Planning
xii. Research and Development
xiii. Institutional Structure (Including Infrastructure)
xiv. Human Resources and Institutional Capacity
xv. Regional and International Cooperation
xvi. Ecosystem Health

The usefulness or application of the development of the NSAAH has now expanded to fit as a critical component of the Progressive Management Pathway for Improving Aquaculture Biosecurity (PMP/AB). The PMP/AB was developed by FAO and partners to guide countries towards achieving sustainable aquaculture biosecurity and health management systems through a bottom-up approach, with strong stakeholder involvement to promote the application of risk management at sector and national levels. The PMP/AB is an extension of the "Progressive Control Pathways" (PCP) stepwise approach which has been internationally adopted to assist countries in developing and monitoring national strategies for the reduction, elimination and eradication of important transboundary diseases of livestock and zoonotic diseases (FAO, 2018, 2019, 2020b). Whereas most PCPs focus on control of single diseases or disease complexes, the PMP/AB focuses on building resilience to aquaculture biosecurity vulnerabilities (i.e. threats), which includes pathogens, poor management practices and lack of capacity in public and private institutions. The PMP/AB is risk-based, proactive and collaborative. It is intended to be flexible and can be applied by any country to manage risks in any aquaculture sector, no matter the current national approach for aquaculture biosecurity in place. The pathway is comprised of four stages and countries decide themselves how far and how fast it is appropriate for them to progress to each stage. Responsibilities must be shared among key national, regional and international stakeholders from government, the production sector and academia, as well as other players in the value chain, building on each other's strengths towards a common goal. The development of a risk-based NSAAH is an important element and end-goal of Stage 1 of the PMP/AB.

7.2 *Problems that need to be addressed*

Outbreaks of TiLVD have resulted in substantial economic losses to producers and pose a threat to food security, especially in under-developed regions. Therefore, information and guidance regarding this disease and its control is urgently needed by both producers and policy makers. Response to the emergence of TiLVD must be rapid. This will require: 1) availability of resources and trained personnel for diagnosis and surveillance; 2) implementation of production management protocols to prevent or control the disease; and 3) development of policies aimed at preventing the disease from spreading among culture facilities or to wild fish populations.

7.3 *Overview of response options*

Responses range from efforts to eradicate the disease from the country to the development of strategies for individual farms to manage production with the TiLV present. The actions are outlined in the following chart (Figure 2). Country-wide eradication strategies (Option 1) can usually only be successful if an initial outbreak of TiLVD is quickly discovered, before it has had a chance to spread among farms or to become established in wild populations. Where eradication is impossible, containment of the disease to certain zones (Option 2) may be possible, especially in countries where production facilities are widely separated. However, if TiLVD becomes either established in wild tilapia populations or widespread among farms, then mitigation through farm management (Option 3) may be the best, or possibly the only, way to reduce losses.

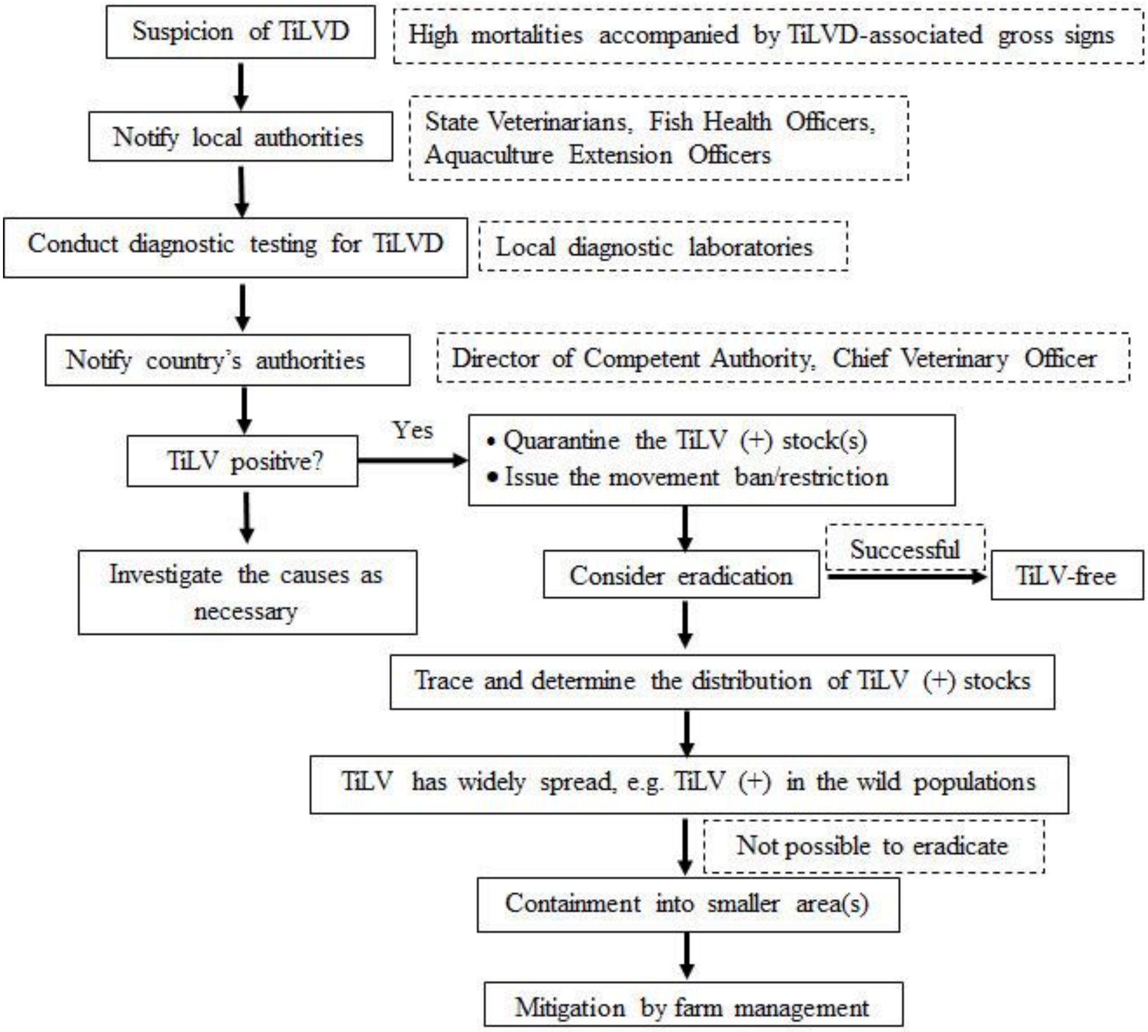

Figure 12. TiLVD responses chart.

7.4 ***Improving knowledge and capability***

A number of international workshops (Table 6) focusing on TiLVD were organized by FAO to provide information and training of aquatic animal health professionals and stakeholders from tilapia producing countries in Asia, the Americas, and Africa.

Table 6. Technical workshops, training courses focused on TiLVD.

Sponsors (Project)	**Title (report website)**	**Location /year**
FAO/China (TCP/INT/SYSU)	Intensive Training Course on Tilapia Lake Virus (TiLV) (http://www.fao.org/fi/static-media/MeetingDocuments/TiLV/Default.html)	China /2018
FAO (GCP/RAF/510/ MUL)	Project Inception Workshop: Enhancing capacity/risk reduction of emerging Tilapia Lake Virus (TiLV) to African tilapia aquaculture (http://www.fao.org/fishery/static/news/TiLVProject/listoflinks.html)	Kenya /2018
FAO (GCP/RAF/510/ MUL)	Enhancing capacity/risk reduction of emerging Tilapia Lake Virus (TiLV) to African tilapia aquaculture (http://www.fao.org/fi/static-media/MeetingDocuments/TiLV/dec2018/Default.html)	Kenya /2018
FAO (UTF/ZAM/077/ ZAM)	Technical Assistance to the Zambia Aquaculture Enterprise: Training Course on Development of an Active Surveillance for Epizootic ulcerative syndrome (EUS) and Tilapia lake virus (TiLV) using the FAO 12-point surveillance checklist (for non-specialists) and its implementation (http://www.fao.org/fi/static-media/MeetingDocuments/TiLV/Presentations/Default.html) Prospectus: (http://www.fao.org/fishery/static/news/prospectus13October2019.pdf)	Zambia /2019

The output of these workshops included:

- development of a surveillance case definition for TiLVD;
- training of fish health-related personnel in diagnosis and surveillance of TiLVD;
- establishment of extension service in fish health;
- TiLVD-targeted surveillance of tilapia farms in several Africa countries, including Angola, Ghana, Kenya, Nigeria, Uganda, and Zambia;
- improvement of farm biosecurity and establishment of good aquaculture practices through training courses for farmers;
- development of NSAAH strategies to support the establishment and long-term maintenance of aquatic biosecurity; and
- establishment of a knowledge sharing and communication network among African countries.

7.5 *Social and economic effects*

Tilapia aquaculture plays an important role in national economies, especially in developing countries; it supplies high quality protein and nutrients for domestic consumption, generates foreign currency, raises producer’s income, and creates employment opportunities. Tilapia aquaculture also plays a core role in rural development in tropical and subtropical regions.

Aquaculture is important for alleviating poverty, reducing hunger and malnutrition, and improving employment, in regions with increasing populations and limited land resources.

TiLVD is one of the major factors constraining the development and expansion of tilapia aquaculture. Government programs are necessary to protect, and to assist individuals and communities most at risk from the disease outbreaks. Especially at risk are those whose livelihoods are largely dependent on fish aquaculture. TiLVD outbreaks can result in dramatic, direct losses both to producers and to local economies. In addition, less obvious effects may be seen such as affected producers being unable to continue paying for pond or land leases or losing the ability to repay loans that were taken to enter fish farming. All of these can negatively impact rural development.

7.6 ***Funding and compensation***

The substantial, financial, cost of the response to a TiLVD outbreak may be borne by:
1) individual producers; 2) tilapia grower associations; 3) aquaculture insurance; 4) regional agencies; 5) national governments; or 6) combinations of these and other stakeholders.

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APPENDIX 1

Group photographs

Intensive training course on tilapia lake virus (TiLV) held at SYSU, Guangzhou, China during 18–24 June 2018.

Project Inception Workshop: Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture held in Nairobi, Kenya during 23–24 October 2018.

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Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture held in at KMFRI, Kisumu, Kenya, during 04–13 December 2018.

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Training course on development of an active surveillance for epizootic ulcerative syndrome (EUS) and tilapia lake virus (TiLV) using the FAO 12-point surveillance checklist (for non-specialists) and its implementation held at UNZA, Lusaka, Zambia during 11–17 October 2019.

This manual provides information to aid the development of contingency plans, of countries, producers, and other stakeholders, to rapidly respond to outbreaks of tilapia lake virus disease (TiLVD). It includes current information relevant to reducing the risk of, and economic losses from, outbreaks of TiLVD. The contents include current knowledge of tilapia lake virus (TiLV) and its hosts; diagnostic methods; epidemiology of the disease; disease prevention and management; planning of national strategies for aquatic animal health (NSAAH) and biosecurity implementation; and potential economic impacts from the disease.

The manual is based, in large part, on information presented and discussed at four FAO training courses/workshops: 1) FAO/China project TCP/INT/SYSU: Intensive training course on tilapia lake virus (TiLV); 2) Project Inception Workshop of GCP/RAF/510/MUL: Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture, held in Nairobi, Kenya; 3) GCP/RAF/510/MUL: Enhancing capacity/risk reduction of emerging tilapia lake virus (TiLV) to African tilapia aquaculture: Intensive training course on TiLV; 4) UTF/ZAM/077/ZAM: Technical assistance to the Zambia Aquaculture Enterprise: Training course on development of an active surveillance for epizootic ulcerative syndrome (EUS) and tilapia lake virus (TiLV) using the FAO 12-point surveillance checklist (for non-specialists) and its implementation.

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